

**Evaluating Self-Attention Placement in R3GAN for Fixed-Scale Single-Image
Super-Resolution: Ablation and Comparative Analysis in Virtualized Single-GPU
Environment**

A Thesis by

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DEDICATION

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TABLE OF CONTENTS

APPROVAL SHEET	ii
DEDICATION	iii
ACKNOWLEDGEMENT	iv
LIST OF FIGURES	ix
LIST OF TABLES	xi
ABSTRACT	xii
CHAPTER 1. INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	3
1.3 Objective of the Study	4
1.4 Hypothesis of the Study	5
1.5 Significance of the Study	5
1.6 Scope and Limitation of the Study	6
CHAPTER 2. REVIEW OF RELATED LITERATURE	8
2.1 Theoretical Framework	8
2.1.1 Generative Adversarial Networks (GANs) Theory	8
2.1.2 R3GAN	9
2.1.3 Self-Attention Mechanism (Non-local Block)	10
2.1.4 Models with Self-Attention Mechanism	11
2.1.5 Super-Resolution Generative Adversarial Networks (SRGAN) . .	12
2.1.6 Training under Constrained Environment	13

2.2	Introduction	13
2.3	Evolution of GAN Architecture	14
2.4	Conditional and Image-to-Image GANs	17
2.5	Attention Mechanisms in GANs	19
2.6	R3GAN as a Modern GAN Baseline	20
2.7	Advances in Image Enhancement and Super-Resolution with GANs	21
2.8	Gaps in the Literature	22
CHAPTER 3. METHODOLOGY		25
3.1	Introduction	25
3.2	Research Design	26
3.2.1	Experimental Variables	26
3.2.2	Experimental Structure	26
3.2.3	Experimental Procedure	27
3.2.4	Robustness and Control Measures	28
3.2.5	Evaluation and Statistical Analysis	28
3.2.6	Scope and Design Limitations	28
3.3	Research Environment	28
3.3.1	Hardware Configuration	29
3.4	Flickr-Faces-HQ Dataset (FFHQ)	31
3.5	Data Preprocessing Steps	32
3.6	Model Architecture	32
3.6.1	Baseline R3GAN Architecture	33
3.6.2	Self Attention Module	36
3.6.3	Self-Attention R3GAN Variants	36
3.6.4	SwinIR and REAL-ESRGAN	40
3.7	Training Strategy	42
3.7.1	Overall Training Framework	42

3.7.2	Loss Function Formulation	43
3.7.3	Optimization Configuration	45
3.7.4	Exponential Moving Average (EMA)	45
3.7.5	Training Procedure	46
3.7.6	Controlled Variant Training	46
3.7.7	Training Stability and Monitoring	47
3.7.8	Implementation Details	47
3.8	Evaluation metrics	48
3.8.1	Peak Signal-to-Noise Ratio (PSNR)	48
3.8.2	Structural Similarity Index Measure (SSIM)	48
3.8.3	Fréchet Inception Distance (FID)	49
3.8.4	Survey-Based Mean Opinion Score (MOS)	49
3.8.5	Blind Subjective Ranking Evaluation	49
CHAPTER 4.	RESULTS AND DISCUSSION	51
4.1	Introduction	51
4.2	Evaluation Context	51
4.3	Hardware Utilization During Training	51
4.4	Ablation Study of Self-Attention Placement	52
4.4.1	Baseline R3GAN as SR	52
4.4.2	Super-Resolution Self-Attention R3GAN	54
4.4.3	Impact of Self-Attention Placement	60
4.5	Selection of the Optimal R3GAN Variant	61
4.5.1	Summary of Model Configuration	61
4.5.2	Quantitative Evaluation Results among SR-SAR3GAN	64
4.5.3	Metric Trade-off Analysis and Decision Criteria	67
4.5.4	Selection of the Optimal R3GAN Variant	67
4.6	Comparative Evaluation Results	68

4.6.1	Metric Results and Analysis	68
4.6.2	Blind Ranking Evaluation	69
4.6.3	Statistical Analysis	70
CHAPTER 5.	SUMMARY	72
5.1	Summary of Contributions	72
5.2	Findings	72
5.3	Limitations	74
5.4	Recommendations	74
REFERENCES		76
APPENDICES		80
	Appendix A. Overall Similarity Result	80
	Appendix B. AI Detection Result	81
	Appendix C. Deed of Assignment	82
	Appendix D. Certification of Substantial Use of University Resources	84

LIST OF FIGURES

2.1	Overview of the Theoretical Framework.	8
3.1	Overview of the Methodology.	25
3.2	FFHQ dataset samples.	31
3.3	Left: downscaled input. Right: 256×256 ground truth.	32
3.4	General overview of our modified R3GAN architecture process.	33
3.5	Illustration of the architecture for image generation using R3GAN.	33
3.6	Structure of the Residual Block, Generator Head, and Discriminator Head.	34
3.7	The concept of the non-local block integrated into the G and D.	36
3.8	The architectural configuration of Self-Attention R3GAN Variant 1.	38
3.9	The architectural configuration of Self-Attention R3GAN Variant 2.	39
3.10	The architectural configuration of Self-Attention R3GAN variant 3.	40
3.11	The architecture of REAL-ESRGAN (Wang et. al., 2021).	41
3.12	Architecture of SwinIR (Liang et. al, 2021)	41
4.1	G and D loss over the training steps.	52
4.2	R1 and R2 gradient regularization values over 65,000 training steps.	53
4.3	G and D loss of SR-SAR3GAN-1.	54
4.4	R1 and R2 regularization of SR-SAR3GAN-1.	55
4.5	G and D loss of SR-SAR3GAN-2.	56
4.6	R1 and R2 regularization of SR-SAR3GAN-2.	57
4.7	G and D loss of SR-SAR3GAN-3.	58
4.8	R1 and R2 regularization of SR-SAR3GAN-3.	59
4.9	The FID over training steps for each model variant and the baseline.	60
4.10	Sample outputs of all developed models compared with the ground truth.	65
4.11	The output of SR-SAR3GAN-2 (left) together with the ground truth (right).	66

4.12	Summary of blind ranking evaluation.	69
4.13	Sample survey outputs: SR-SAR ₃ GAN-2, Real-ESRGAN, SwinIR.	70

LIST OF TABLES

3.1	Research Environment Hardware and Software Configuration	30
4.1	Training summary of all SR-SAR ₃ GAN variants.	61
4.2	Quality metrics for the performance across all four model variants. . .	64
4.3	Objective metrics (FID, SSIM, PSNR) and survey-based MOS.	68
4.4	Friedman and Wilcoxon tests for perceptual ranking results.	70
4.5	Summary of SR model experiments.	71

ABSTRACT

R3GAN has recently demonstrated strong potential as a simplified generative adversarial network (GAN) baseline capable of achieving competitive image quality relative to modern GAN architectures and transformer-based models. This study adapts R3GAN for single-image fixed-scale super-resolution and systematically evaluates the impact of incorporating self-attention mechanisms and their placement on perceptual quality. Four architectural variants were developed: a baseline super-resolution R3GAN without self-attention and three self-attention-enhanced variants with attention modules inserted at different network stages. Experimental results indicate that mid-level self-attention placement yields the best overall performance, with FID 7.42 and MOS 3.94/5, improving over the baseline and matching models like ESRGAN and SwinIR when trained under the same single-GPU environment. Additionally, the findings exhibited a divergence between objective reconstruction metrics and human perception: SwinIR achieves higher pixel-wise accuracy (PSNR of 30.53 dB) but lower subjective score (MOS of 2.88) than SRSAR3GAN-2, indicating better reconstruction fidelity does not guarantee better perceived quality. Statistical analysis confirms that SRSAR3GAN-2 achieves perceptual quality comparable to established benchmarks. The results establish R3GAN as a viable modern baseline for generative super-resolution research and demonstrate that self-attention placement plays a critical role in balancing perceptual quality and computational efficiency. This shows that competitive super-resolution performance can be achieved within a Virtualized single-GPU training infrastructure.

Keywords: Single-Image Super-Resolution, R3GAN, Self-Attention Mechanism, Virtualized Single-GPU Training

CHAPTER 1. INTRODUCTION

1.1 Background of the Study

Since Ian Goodfellow and his team at the University of Montreal develop the Generative Adversarial Network (GAN)s in 2014, the field of generative image synthesis has witnessed rapid growth and progress. It is now a baseline model not only for generating synthetic data but also utilized for other task in various fields, example of this are image enhancement and restoration. GANs work through the use of two neural networks, a generator and a discriminator, this networks then engaged in a minimax game configuration, which will results to the creation of synthetic data that tends to conform to the actual data distribution. This simple mechanics allowed Generative Adversarial Networks to be applied to other tasks example of this are image to image translation, image restoration, and super resolution.

Despite these notable achievements, early GAN models suffered from various issues, including training instability, mode collapse, and vanishing gradients, this are largely due to the adversarial optimization process. To address these issues, multiple GAN variants emerged that incorporate architectural constraints, alternative loss functions, and improved training strategies, These developments have resulted in significant improvements in synthesis quality, convergence stability, and model control, allowing GANs to generate high quality and realistic outputs.

The recent innovations in GANs have explored various directions including larger models, novel architectures, and enhanced training techniques, a parallel research direction surfaces that questions whether architectural complexity is always necessary to improve the GAN architecture. Example of this research is R3GAN, a minimalist GAN architecture that streamlines the StyleGAN2 backbone and introduces a relativistic regularization loss to improve stability and convergence (Huang et al., 2025). Despite its reduced architectural complexity, R3GAN demonstrated competitive or superior

performance when compared to more complex models on standard image generation benchmarks. To build on this foundation, prior works have shown that self-attention mechanisms can significantly improve image generation and restoration quality by enabling models to capture long-range dependencies and global contextual relationships. This capability was notably shown in Self-Attention GAN (SAGAN) where attention-based feature interactions enhanced structural coherence and detail preservation for generated outputs (Zhang et al., 2018).

Building on this foundation, prior work has shown that self-attention mechanisms can significantly improve image generation and restoration quality by enabling models to capture long-range dependencies and global contextual relationships. This capability was notably demonstrated in Self-Attention GAN (SAGAN), where attention-based feature interactions enhanced structural coherence and detail preservation (Zhang et al., 2018). However, R3GAN in its original formulation does not incorporate self-attention mechanisms, nor has it been extensively explored for conditional image-to-image tasks such as single-image super-resolution.

In addition, most GAN-based super-resolution models are commonly trained and evaluated using high-end computational resources. Their performance, stability, and practicality under resource-constrained environments remain underexplored, despite the growing need for efficient and deployable super-resolution systems. In response to these gaps, this study reformulates R3GAN as a conditional single-image super-resolution model, where low-resolution inputs of 64×64 pixels are enhanced to 256×256 high-resolution, and integrates self-attention mechanisms within its architecture. The proposed approach evaluates the impact of self-attention placement on super-resolution quality, training stability, and computational efficiency under limited computational resources, thereby extending the applicability of R3GAN to practical, constrained super-resolution scenarios.

1.2 Statement of the Problem

Although R3GAN demonstrates improved training stability and image generation quality through architectural simplification and regularization, its effectiveness for single-image super-resolution under constrained conditions remains largely unexplored. In particular, R3GAN has not been evaluated for conditional super-resolution tasks due to the previous implementation being a generative GAN model.

Furthermore, R3GAN does not incorporate self-attention mechanisms, which have been shown to improve global feature modeling and perceptual quality in both GAN-based and Transformer-based super-resolution models (Zeng et al.,2024). The absence of attention limits its ability to capture long-range spatial dependencies that are critical for recovering fine structural details in high-resolution outputs.

This study addresses these gaps by integrating self-attention mechanisms into the R3GAN architecture and adapting it for conditional single image super-resolution under resource-constrained training environments:

1. Can the adapted R3GAN produce visually enhanced images with great detail and realism when trained on a virtualized single GPU environment?
2. How does the integration of self-attention mechanism into the R3GAN architecture affect its performance in enhancing images when trained under limited computational resources?
3. Does the self-attention-enhanced R3GAN produce visually improved images with better perceptual quality and preserved details compared to a conditional R3GAN-based super-resolution model without self-attention under the same resource constraints?
4. What is the impact of self-attention on training stability, convergence behavior, and computational efficiency within the R3GAN architecture in a single-GPU

setting?

5. How does the performance of the modified R3GAN compare with state-of-the-art image enhancement models such as SwinIR and REAL-ESRGAN in terms of image quality and perceptual metrics trained under the same environment?

1.3 Objective of the Study

The primary objective of this study is to adapt and enhance the R3GAN architecture by integrating self-attention mechanisms and reformulating it as a conditional fixed-scale single-image super-resolution model. The modified R3GAN was trained and evaluated under resource-constrained computational conditions to analyze its stability, efficiency, and effectiveness in generating 256×256 high-resolution images from 64×64 low-resolution inputs.

Specific Objectives:

1. Modify the R3GAN architecture by integrating self-attention mechanisms and converting it from an unconditional generator into a conditional fixed-scale single-image super resolution framework.
2. Train and evaluate the modified R3GAN in a virtualized-training environment to examine its training stability and efficiency.
3. Compare the impact of self-attention on super-resolution performance in terms of perceptual quality, structural consistency, and visual detail for 256×256 resolution outputs.
4. Quantify the effect of self-attention on training stability, convergence behavior, and computational efficiency within the R3GAN architecture under the virtualized-

training environment.

5. Identify the most effective placement of self-attention mechanism within the R3GAN architecture that balances image enhancement quality and computational efficiency under resource-constrained conditions.
6. Conduct a comparative analysis of the modified R3GAN and the two leading image enhancement models: REAL-ESRGAN and SwinIR.

1.4 Hypothesis of the Study

Null Hypothesis (H_0): The placement of self-attention mechanisms within the R3GAN architecture for single-image super-resolution produces no significant performance differences across configurations, and no configuration achieves image quality comparable to state-of-the-art models when trained on virtualized single-GPU infrastructure.

Alternative Hypothesis (H_1): Strategic placement of self-attention mechanisms within the R3GAN architecture significantly impacts super-resolution performance, with optimal placement configurations achieving image quality comparable to baseline R3GAN and state-of-the-art models quantitative (per quantitative and qualitative metrics) while enabling feasible training on virtualized single-GPU infrastructure.

1.5 Significance of the Study

This study contributes to the ongoing advancement of generative adversarial networks (GANs) by enhancing the R3GAN architecture through the integration of self-attention mechanisms and its adaptation into a conditional fixed-scale single-image

super-resolution model. R3GAN was selected as the foundation due to the author's claims of the viability of R3GAN as a new GAN baseline along with strong convergence stability, and effective balance between model simplicity and output quality (Huang et al., 2025). The inclusion of self-attention mechanism to the model is expected to improve and better capture global image dependencies and preserve fine-grained details often lost in convolution-based approaches. Furthermore, training the modified R3GAN in a constrained-resource computational environment provides insights into its adaptability, efficiency, and scalability beyond high-end hardware outside unconditional task. These findings benefit researchers and practitioners in optimizing GANs for resource-constrained real-world applications and inform future lightweight, attention-driven generative models to enhance accessibility and deployment of image enhancement technologies.

1.6 Scope and Limitation of the Study

The study focuses on exploring the capabilities of the R3GAN architecture by integrating self-attention mechanism specifically non-local block variant and adapting it as a conditional fixed-scale single-image super-resolution model, where low-resolution inputs of 64×64 pixel images were enhanced to 256×256 resolution. This refinement was made to ensure stable training, controlled experimentation, and feasibility within the available computational resources. The modified model was trained and evaluated in a limited-resource computational environment to assess its performance, efficiency, and adaptability under these conditions.

The scope of the study is only limited to ablation-based analysis and comparative analysis. The ablation of this study was conducted to examine the effects of incorporating self-attention into the R3GAN architecture by comparing the self-attention-enhanced model against the baseline R3GAN under the same experimental settings. While

the comparative analysis was performed against established state-of-the-art image super-resolution models, specifically REAL-ESRGAN and SwinIR, to contextualize the proposed model's performance in terms of image quality and perceptual metrics.

The study is limited to static image enhancement only and did not address anything related to video processing, real-time inference, or higher-resolution image generation beyond the fixed-scale input of 64x64 to 256x256 resolution. All of the experiments were conducted using a single dataset, which may have limited the generalizability of the findings to other datasets or domains. In addition to this, due to hardware, computational constraints, model training duration and hyperparameter tuning may have been affected, potentially influencing the comparative results. The comparative evaluation is intended to provide performance context to the adapted architecture rather than exhaustive benchmarking, with emphasis placed on architectural analysis and efficiency within a low-resource setting.

the following value function:

minimize over G , maximize over D :

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad \text{Equation (1)}$$

In this equation, x denotes samples from the real data distribution p_{data} , $z \sim p_z$ represents samples from a noise prior, $G(z)$ is the synthetic sample generated from the noise z , and $D(\cdot)$ outputs the probability that the input is real.

Proof of Optimality:

Goodfellow et al. showed that for a fixed generator G , the optimal discriminator D^* is:

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)} \quad \text{Equation (2)}$$

where p_g is the distribution of generated samples. The global optimum is reached when $p_g = p_{\text{data}}$, meaning the generator perfectly models the real data distribution. This demonstrates GAN's theoretical ability to learn complex data distributions.

2.1.2 R3GAN

R3GAN improves classical GANs by introducing the relativistic discriminator, which estimates the probability that a real data sample is more realistic than a fake one rather than independently classifying samples. This relativistic approach modifies the discriminator output as:

$$D_{\text{rel}}(x, x') = \sigma(C(x) - C(x')) \quad \text{Equation (3)}$$

Here, $C(\cdot)$ is the critical function (logits before the sigmoid), σ is the sigmoid

function, x is a real sample, and x' is a generated sample. By comparing real and fake samples, the relativistic discriminator encourages the generator to produce outputs that are more realistic relative to the real data distribution.

The loss function for R3GAN consists of a relativistic adversarial loss and regularization terms that stabilize training and improve convergence.

The relativistic adversarial discriminator loss is defined as:

$$\mathcal{L}(\theta, \psi) = \mathbb{E}_{\substack{x \sim p_z \\ x \sim p_D}} \left[f(D_\psi(G_\theta(z)) - D_\psi(x)) \right] \quad \text{Equation (4)}$$

where $C(\cdot)$ denotes the discriminator logits (pre-sigmoid), x is a real sample, and $G(z)$ is a generated sample from noise z .

Additional gradient penalty terms, R_1 and R_2 regularizations, are added to stabilize the discriminator:

$$R_1(\psi) = \frac{\gamma}{2} \mathbb{E}_{x \sim p_{\text{data}}} \left[\|\nabla_x D_\psi(x)\|_2^2 \right], R_2(\theta, \psi) = \frac{\gamma}{2} \mathbb{E}_{x \sim p_\theta} \left[\|\nabla_x D_\psi(x)\|_2^2 \right] \quad \text{Equation (5)}$$

Finally, the total R3GAN loss combines the adversarial and regularization terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}(\theta, \psi) + R_1(\psi) + R_2(\theta, \psi) \quad \text{Equation (6)}$$

Empirical results and theoretical justification in Jolicoeur-Martineau (2018) demonstrate that this relativistic formulation improves training stability and encourages the generator to produce outputs that better match the real data distribution.

2.1.3 Self-Attention Mechanism (Non-local Block)

Traditional convolutional layers focus on neighboring pixels, which inherently limits the model's ability to capture long-range dependencies across an image. The Self-Attention mechanism (Non-local Block) solves this by letting each location in the image

look at every other location, then combining all that information using learned weights. This is achieved by transforming the input embeddings into three distinct feature spaces: Queries (Q), Keys (K), and Values (V) via 1×1 convolutions, where Q and K are projected to $C/8$ channels for computational efficiency, while V retains the full C channels. Attention scores are computed from these projections to determine how much focus each position places on every other position, enabling all locations to be processed simultaneously in parallel. The fundamental operation is defined as:

$$Y_i = \frac{1}{C(\mathbf{X})} \sum_{\forall j} f(\mathbf{X}_i, \mathbf{X}_j) g(\mathbf{X}_j) \quad \text{Equation (7)}$$

where $\mathbf{X}$ is the input feature map, and i, j are spatial positions. The score $f(\mathbf{X}_i, \mathbf{X}_j)$ is computed as the dot product between the Query at position i and the Key at position j , normalized via Softmax to produce the attention map. The function $g(\mathbf{X}_j)$ represents the Value projection. The attended output is then scaled by a learnable parameter γ and added back to the input as a residual connection:

$$\hat{Y} = x + \gamma \cdot \text{Attention}(Q, K, V) \quad \text{Equation (8)}$$

By attending to all positions, the model captures global context and preserves structural details.

2.1.4 Models with Self-Attention Mechanism

Multiple GANs have already integrated self-attention in their architecture to enhance their ability to model long-range spatial dependencies and global contextual relationships. Self-Attention Generative Adversarial Networks (SAGAN) introduced by Zhang et al. (2019), which integrates self-attention layers into both the generator and the discriminator to model long-range spatial dependencies. TransGAN, proposed by Jiang et al. (2021), replaces convolutional layers entirely with Transformer blocks

composed of multi-head attention, demonstrating that attention-based architectures alone can effectively support adversarial image generation. Meanwhile, U-GAT-IT (Kim et. al., 2020) integrates self-attention mechanisms with adaptive normalization techniques to enhance image-to-image translation. These studies show the effectiveness of integrating self-attention to improve global feature modeling.

2.1.5 Super-Resolution Generative Adversarial Networks (SRGAN)

Super-Resolution Generative Adversarial Networks (SRGAN), introduced by Ledig et al. (2017), extend the GAN framework to the task of single-image super-resolution by learning a mapping from low-resolution images to their high-resolution counterparts. Unlike traditional pixel-wise regression approaches, SRGAN employs adversarial learning to encourage perceptually realistic high-frequency detail generation.

In SRGAN, the generator is trained to produce super-resolved images that are indistinguishable from real high-resolution images, while the discriminator learns to differentiate between generated and ground-truth high-resolution samples. To improve perceptual quality, SRGAN incorporates a perceptual loss function, which combines adversarial loss with content loss computed in a high-level feature space extracted from a pretrained convolutional network.

Formally, the super-resolution objective can be expressed as:

$$\begin{aligned} \min_{\theta_G} \max_{\theta_D} \mathbb{E}_{I^{HR} \sim p_{\text{train}}(I^{HR})} [\log D_{\theta_D}(I^{HR})] \\ + \mathbb{E}_{I^{LR} \sim p_G(I^{LR})} [\log (1 - D_{\theta_D}(G_{\theta_G}(I^{LR})))] \end{aligned} \quad \text{Equation (9)}$$

SRGAN established adversarial learning as an effective paradigm for super-resolution by prioritizing perceptual realism over purely distortion-based metrics. However, its training stability and computational demands remain challenging, particularly under limited-resource settings. These limitations motivate the exploration of more efficient GAN variants, such as R3GAN, and the integration of self-attention mechanisms to

enhance global feature modeling in super-resolution tasks.

2.1.6 Training under Constrained Environment

Modern AI training often uses large-scale systems with multiple GPUs, high-end hardware, and long training times (Shen et al., 2024). Virtualized GPU environments offer an alternative, giving access to similar hardware but with limits such as single-GPU use, 5–15% overhead, and variable performance.

Training deep learning models in virtualized environments necessitates careful architectural and methodological considerations. Models must balance performance with computational feasibility, as multi-GPU scaling and distributed training are often unavailable. Virtualization-induced variability requires robust training strategies including deterministic CUDA operations, conservative batch sizing, and comprehensive checkpointing. While deterministic settings can be enforced, hardware-level variations and virtualization layer interactions may introduce minor numerical differences that must be distinguished from architecturally significant performance variations (Chen & Wen, 2022; Semmelrock et al., 2025).

This study uses a single virtualized GPU (NVIDIA V100) to evaluate GAN-based super-resolution under these constraints. It aims to relate methods developed on large systems to setups with limited resources. This reflects typical use in education, small labs, and cloud workflows, where efficiency and stable optimization are prioritized over scale.

2.2 Introduction

Generative adversarial networks (GANs) learn data distributions to generate realistic images (Goodfellow et al., 2014; Huang et al., 2025). They were first defined as a minimax game between a generator and a discriminator, allowing image generation in

one forward pass. However, early GANs were difficult to train and prone to instability and mode collapse (Arjovsky et al., 2017).

Subsequent work improved GANs through better loss functions, normalization, and architectures. GANs are now used for image enhancement and structured generation. Conditional GANs (cGANs) extend this by learning mappings from input images to outputs, supporting tasks like style transfer, super-resolution, and denoising (Mirza & Osindero, 2014; Isola et al., 2017; Zhu et al., 2017; Ledig et al., 2017).

Pix2Pix and CycleGAN showed that GANs can learn mappings from paired or unpaired data. GAN-based super-resolution and denoising improve perceptual quality over standard convolutional methods.

Attention mechanisms are also used in GANs. Self-attention models relationships between distant spatial regions. SAGAN (Zhang et al., 2019) added multi-head self-attention to GANs to model long-range dependencies. Later work improved attention-based GANs for better image quality (Xu et al., 2018; Jiang et al., 2021; Hudson & Zitnick, 2021; Zhang et al., 2022).

2.3 Evolution of GAN Architecture

Since the original GAN formulation (Goodfellow et al., 2014), multiple variants have emerged, each addressing specific shortcomings or adding new capabilities. Early work introduced practical remedies to training instability and mode collapse. For instance, the Deep Convolutional GAN (DCGAN) (Radford et al., 2016) established architectural guidelines for convolutional GANs, using batch normalization and avoiding pooling to stabilize training. Incorporated convolutional architectures with batch normalization and ReLU activations, enabling more stable training on image data. Demonstrated that GANs can learn good feature representations in unsupervised settings.

Wasserstein GAN (WGAN) (Arjovsky et al., 2017) Replaced the original GAN loss

with the Wasserstein distance, providing meaningful gradients even when distributions do not overlap. The Wasserstein GAN (WGAN) reduced mode collapse by penalizing the Earth Mover's distance. Their follow-up (WGAN-GP) by Gulrajani et al.(2017) added a gradient penalty instead of weight clipping, yielding smoother, more stable convergence. Other loss variants included the Least Squares GAN (LSGAN), which uses L2 loss (Mao et al., 2017) and the Relativistic GAN (Jolicoeur-Martineau, 2019), which compares real and fake logits. Spectral normalization (Miyato et al., 2018) was introduced to constrain the discriminator's Lipschitz constant, further preventing exploding gradients.

Spectral Normalization (SNGAN) Enforced a unit spectral norm on discriminator weights, controlling its capacity and improving training stability. SNGAN demonstrated state-of-the-art image quality on CIFAR-10 and ImageNet with minimal additional overhead (Miyato et al., 2018). Self Attention GAN (SAGAN) Introduced multi-head self-attention in both the generator and discriminator, allowing the model to capture long-range dependencies. SAGAN achieved a dramatic jump in Inception score (from 36.8 to 52.5 on ImageNet) and reduced FID, illustrating the benefit of global context for fine image details (Zhang et al., 2018). BigGAN scaled up GANs to generate high-fidelity, large-scale images. By increasing model and batch size and using class-conditioning with projection, BigGAN produced class-conditional ImageNet images at 512x512 with unprecedented quality(Brock et al., 2019) highlighting the importance of scale and class information for GAN fidelity.

Generators that grow images from low to high resolution progressively were implemented by Progressive GAN in which it adds new layers during training. This architectural strategy enabled stable training at resolutions up to 1024x1024 (Karras et al.2017), laying the groundwork for ultra-high-resolution synthesis. StyleGAN proposed a style-based generator where latent codes are first mapped to "style" vectors that modulate convolutional layers. This architecture allows disentangled control over

image features (e.g. coarse geometry, fine texture) and produces highly realistic faces (Karras et al., 2019). Its follow up in 2020 StyleGAN2, introduced by Karras et al. in 2020, addressed several limitations observed in the original StyleGAN architecture, particularly the presence of characteristic visual artifacts often described as “water-droplet” distortions.

These artifacts arose due to how the network synthesized images, especially in the way style information was injected into the generation process. To combat this, the authors introduced weight demodulation, a technique that replaced the instance normalization used in StyleGAN, effectively normalizing feature maps on a per-channel basis to reduce feature entanglement and improve the control over style. Additionally, they implemented path length regularization, a training technique designed to encourage a more consistent mapping between the latent space and the output image space.

This regularization penalized deviations in the image’s sensitivity to changes in latent codes, promoting smoother and more stable image synthesis. Together, these enhancements not only eliminated many of the previous artifacts but also significantly improved the overall fidelity and quality of generated images, while making the training process more stable and predictable. StyleGAN3 then addressed aliasing in generators by ensuring continuous transformations of generated images (alias-free design). This enabled spatial disentanglement (e.g. image could be translated/rotated smoothly) and prevented artifacts when generating high-resolution outputs (Karras et al., 2021).

Recent work has revisited GANs in the era of massive datasets and computation. GigaGAN (Kang et al., 2023) demonstrates that with new architectures and training strategies, GANs can be scaled to text-to-image generation on millions of data samples, rivaling diffusion models in both speed and quality. StyleGAN-T (Sauer et al., 2023) adapts StyleGAN for fast large-scale text-to-image synthesis. Both works highlight that GANs remain competitive when carefully scaled, contrary to the belief that diffusion models have entirely taken over (Kang et al., 2023 & Sauer et al., 2023).

Throughout this evolution, each variant targeted known GAN problems: mode collapse, unstable gradients, or lack of control. For example, different loss functions (WGAN, WGAN-GP, LSGAN) tackled stability; normalization schemes (spectral norm, batchnorm) controlled divergences; and architecture changes (progressive growing, style-based layers) increased fidelity and user control. Collectively, these advances have dramatically increased GAN performance and usability. In particular, the search for more stable and scalable GANs culminates in the recent R3GAN (Huang et al., 2025).

2.4 Conditional and Image-to-Image GANs

While many early GAN variants focused on unconditional image generation, a wide range of practical computer vision tasks require learning explicit mappings between images, such as translating low-quality images to high-quality ones or transforming images between visual domains. Conditional Generative Adversarial Networks (cGANs), introduced by Mirza and Osindero (2014), provided a framework for such problems by conditioning both the generator and discriminator on auxiliary information y , such as class labels or input images. This conditioning allows controlled generation, where the output is guided by the input.

In their foundational work, Mirza and Osindero demonstrated that by feeding class labels to both the networks, a GAN can generate images belonging to a specified class, as shown in their experiments on MNIST digit generation. This shows that conditioning can control multi-modal outputs using input variables, extending GANs beyond unconditional generation.

A key class of conditional GANs is image-to-image translation, where the condition y is an image. Pix2Pix (Isola et al., 2017) introduced a supervised method for paired data, learning a mapping $G: X \rightarrow Y$ between input and target images. It uses a U-Net generator and PatchGAN discriminator, learning both the mapping and loss. It was applied to

tasks such as edge-to-image translation, label-to-image synthesis, and colorization.

CycleGAN (Zhu et al., 2017) extended image-to-image translation to unpaired data using cycle-consistency loss, enabling training without paired images. By learning two inverse mappings and enforcing $F(G(X)) \approx X$, CycleGAN demonstrated successful domain translation across visually distinct domains such as horses and zebras or summer and winter scenes. Subsequent extensions such as Pix2PixHD and SPADE (Park et al., 2019) further improved conditional generation quality, enabling high-resolution image synthesis from semantic layouts. Multi-domain translation frameworks like StarGAN (Choi et al., 2018) generalized conditioning further by allowing a single generator to translate among multiple visual domains using domain labels.

Conditional GANs have also been widely applied to image restoration and enhancement tasks, particularly super-resolution. In super-resolution, the conditional input is a low-resolution (LR) image and the desired output is its high-resolution (HR) counterpart. SRGAN (Ledig et al., 2017) was the first to apply GANs to single-image super-resolution, combining adversarial loss with a perceptual loss based on deep VGG features. The authors demonstrated that while pixel-wise optimization metrics such as PSNR produce overly smooth results, adversarial training enables sharper and more realistic textures. REAL-ESRGAN (Wang et al., 2018) improved SRGAN by using Residual-in-Residual Dense Blocks (RRDB), removing batch normalization, and applying a relativistic adversarial loss. These changes improved texture realism and perceptual quality, making it a strong GAN-based super-resolution model.

Beyond super-resolution, conditional GANs are used for denoising, deblurring, and inpainting. DnCNN (Zhang et al., 2017) remains a strong denoising baseline due to its pixel-wise objective. GAN-based methods instead learn a mapping from noisy to clean images, using adversarial loss to improve perceptual realism (Chen et al., 2018; Wang et al., 2019). Similar conditional GAN setups are used for motion deblurring (Kupyn et al., 2018) and inpainting (Pathak et al., 2016).

Conditional GANs provide a general framework for image-to-image tasks, including super-resolution, denoising, and style transfer. However, many approaches rely on older architectures or trade off between perceptual quality and pixel accuracy, without using recent GAN improvements. This study addresses this by adapting R3GAN into a conditional framework and integrating self-attention, aiming to improve stability and image quality for low-resource super-resolution.

2.5 Attention Mechanisms in GANs

Attention mechanisms improve deep learning models by capturing dependencies across distant spatial regions. In GANs, self-attention enables both generator and discriminator to use global context instead of only local receptive fields (Yuan et al., 2020). This is important for high-resolution synthesis and super-resolution, where distant regions must remain consistent.

One of the first GANs to use attention is SAGAN (Zhang et al., 2019). It introduces a self-attention layer that allows features at any spatial position to affect others. Unlike convolution-only GANs, it models long-range interactions, improving global structure and detail. SAGAN achieved strong results on ImageNet, with an Inception Score of 52.5 and FID of 18.65. Its attention maps also show meaningful relationships between object parts.

The mechanism is related to non-local networks (Wang et al., 2018), where each feature is computed as a weighted sum over all positions. This formulation enables long-range dependency modeling and improves structural consistency in tasks like super-resolution.

Attention has also been used in conditional GANs. AttnGAN (Xu et al., 2018) applies attention to align text with image regions during generation. It improves fine-grained control by focusing on relevant words at each stage, showing that attention

strengthens conditioning.

Transformer-based GANs further extend this idea. TransGAN (Jiang et al., 2021) removes convolutions and uses only self-attention over image patches. StyleSwin uses Swin Transformer blocks for high-resolution synthesis. These models improve quality but require high computational cost and large datasets, limiting practical use in low-resource settings.

Overall, attention improves GANs by enabling non-local feature interactions and better global structure. This is most useful in high-resolution tasks. However, full Transformer GANs are expensive, motivating hybrid designs that combine convolution with selective attention.

Despite these advances, attention is still rarely integrated into lightweight GAN baselines. R3GAN (Huang et al., 2025) excludes attention but identifies it as future work. This study integrates non-local self-attention into R3GAN to improve long-range dependency modeling in conditional super-resolution, while maintaining its efficiency and stability.

2.6 R3GAN as a Modern GAN Baseline

Huang et al. (2025) introduced R3GAN as a simplified GAN baseline aimed at revisiting core design choices. Instead of relying on many stabilization tricks, it uses a relativistic discriminator loss with explicit regularization. The method is supported by a local convergence analysis, providing a theoretical basis for stable training (Huang et al., 2025).

R3GAN uses a residual convolutional backbone rather than style-based designs such as StyleGAN. Each resolution stage contains residual blocks with $1\times 1-3\times 3-1\times 1$ convolutions and LeakyReLU activations. It removes batch normalization and stochastic noise inputs. Stability is maintained using Fixup initialization, where the last layer of

each residual block is zero-initialized and other layers are scaled by $1/\sqrt{L}$. Grouped 3×3 convolutions (group size 16) are used to increase channel capacity with lower cost.

The model excludes common GAN components such as progressive growing, style modulation, minibatch standard deviation, and entropy regularization. Despite this, training remains stable without mode collapse. Ablation results show reliable convergence even when auxiliary components are removed. R3GAN achieves strong performance on FFHQ, CIFAR-10, CIFAR-100, and ImageNet, outperforming StyleGAN2 and competing with diffusion-based models, with around 25M parameters (Huang et al., 2025).

The authors note that R3GAN does not include self-attention or Transformer components. They suggest that adding attention blocks or adopting Transformer-based designs could further improve performance.

Overall, R3GAN provides a stable and efficient convolutional GAN baseline. Its lack of attention makes it suitable for extension. This study builds on R3GAN by adding conditional generation and non-local self-attention to improve global context modeling for super-resolution, following the directions proposed by Huang et al. (2025).

2.7 Advances in Image Enhancement and Super-Resolution with GANs

The computer vision community has developed many models for low-level vision tasks such as super-resolution (SR) and denoising. These include both GAN-based and CNN-based methods.

SRGAN introduced adversarial learning for single-image super-resolution by combining GAN loss with perceptual loss from VGG features (Ledig et al., 2017). It showed that optimizing for PSNR and SSIM produces smooth results, while adversarial training improves sharpness and texture, but may reduce distortion metrics.

REAL-ESRGAN extended SRGAN by improving architecture and training design

(Wang et al., 2018). It replaces residual blocks with Residual-in-Residual Dense Blocks (RRDB), removes batch normalization, and uses a relativistic adversarial loss similar to later GAN formulations such as R3GAN. It also refines perceptual loss using pre-activation features. These changes improve texture realism and stability over SRGAN.

CNN-based models remain strong in distortion metrics. RCAN uses deep residual blocks with channel attention to emphasize informative features (Zhang et al., 2018). It achieves high PSNR and SSIM on benchmarks but produces less realistic textures compared to GAN-based methods.

For denoising, DnCNN is widely used due to residual learning and batch normalization, handling multiple noise levels with one model (Zhang et al., 2017). GAN-based denoising methods treat the task as image-to-image translation (Mao et al., 2017; Chen et al., 2018), but usually achieve lower fidelity than CNN-based approaches, despite better visual outputs in some cases.

GANs are also applied to deblurring (DeblurGAN; Kupyn et al., 2018) and inpainting (Pathak et al., 2016; Yu et al., 2018), showing their flexibility in image restoration tasks.

Overall, there is a trade-off between perceptual quality and distortion metrics. GAN-based methods such as REAL-ESRGAN produce sharper textures but may introduce artifacts and lower PSNR, while CNN-based models like RCAN achieve higher fidelity but weaker perceptual realism. This motivates methods that balance both objectives. This study explores a self-attention-enhanced R3GAN for super-resolution to improve global context modeling while maintaining GAN stability.

2.8 Gaps in the Literature

R3GAN (Huang et al., 2025) achieves strong results in unconditional image synthesis by generating images from noise without input guidance. Super-resolution, however, is a conditional task that requires a mapping from a low-resolution (LR) image to

a corresponding high-resolution (HR) output. This difference creates a gap, as the applicability of modern unconditional GANs like R3GAN to supervised super-resolution has not been fully explored.

Adapting unconditional GANs to conditional settings is not straightforward. It requires modifying the generator and discriminator to accept input images and adding reconstruction-based objectives to preserve pixel alignment between LR and HR images. Laria et al. (2022) note that such adaptation requires architectural changes, especially when spatial correspondence is required. This is relevant for applications such as image restoration and enhancement, where output control is necessary (Bourou et al., 2024).

Existing SR methods such as SRGAN (Ledig et al., 2017) use adversarial and pixel-wise losses to model this mapping but rely on older GAN architectures that emphasize local convolutions and limited global context. More recent models like SwinIR use attention mechanisms to improve reconstruction quality but remove adversarial learning and focus on deterministic outputs. This creates a split between GAN-based SR models with weaker global modeling and attention-based models without adversarial training.

This separation highlights a gap in combining modern unconditional GANs with conditional SR tasks. In particular, it is unclear whether stable and high-quality generators such as R3GAN can be adapted effectively to supervised super-resolution. In addition, although self-attention improves performance in SR and restoration tasks, its integration into relativistic GAN frameworks remains limited.

Self-attention enables each spatial position to use information from all others, improving global consistency. SAGAN (Zhang et al., 2019) showed that adding self-attention to GANs improves structure and detail. Recent studies also report that attention improves feature representation in restoration tasks (Ali et al., 2023). However, these methods are rarely combined with relativistic GANs for SR.

This study addresses these gaps by adapting R3GAN to a conditional super-resolution

framework and adding self-attention modules. The goal is to combine (1) the stability of modern GANs, (2) conditional learning for SR, and (3) global context modeling from attention. The focus is not to replace existing SR methods, but to evaluate whether a self-attention-enhanced R3GAN can perform effectively under limited computational resources.

CHAPTER 3. METHODOLOGY

3.1 Introduction

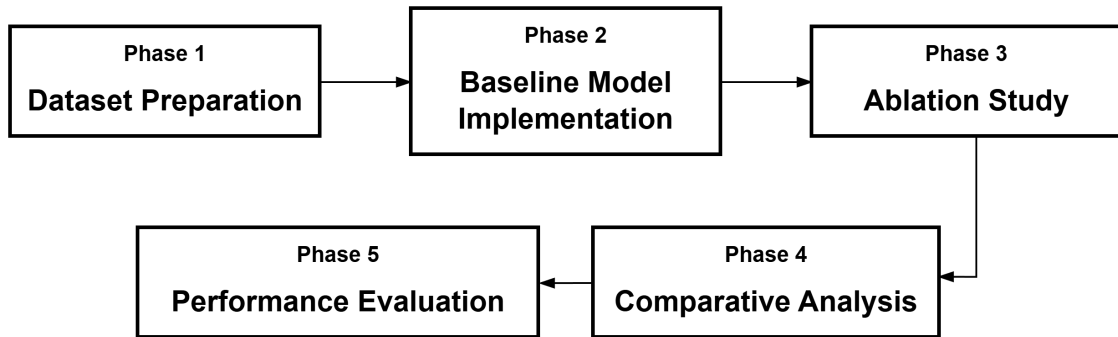


Figure 3.1. Overview of the Methodology.

This chapter presents the methodological framework employed to adapt the R3GAN architecture by integrating a self-attention mechanism and reformulating it as a conditional Generative Adversarial Network (GAN) for image super-resolution tasks. The study used an experimental, quantitative design, combining ablation studies of the architecture with comparative evaluation of the modified R3GAN against existing state-of-the-art super-resolution models.

The experimental framework evaluates how self-attention placement in the R3GAN architecture affects single-image super-resolution. A placement-based ablation was used to find configurations that improve image quality with minimal added computation. This also tests the model’s suitability for resource-limited environments.

In addition to the ablation analysis, the proposed R3GAN variants were evaluated comparatively against established image super-resolution models using standard quantitative metrics and qualitative visual analysis. Computational efficiency was also evaluated under the same low-resource training setup. This chapter describes the experimental workflow, research design, dataset and preprocessing, model architectures, training setup, and evaluation metrics used in the study.

3.2 Research Design

This study uses an experimental, quantitative design to examine how architectural changes affect performance in an adversarial learning setup. It focuses on identifying the impact of specific components through ablation studies under controlled training conditions.

The design has two parts: (1) an ablation study on the effect and placement of self-attention, and (2) a comparison between the best-performing variant and baseline models under the same conditions.

3.2.1 Experimental Variables

Independent Variables

- Presence or absence of a self-attention module
- Placement of self-attention in the generator and discriminator, based on the variants in Section 3.6.3

Dependent Variables

- Image quality metrics: PSNR, SSIM, and FID
- Subjective quality measured using Mean Opinion Score (MOS)
- Training behavior, including convergence and loss stability
- Computational cost, including epoch time, GPU usage, and parameter count

3.2.2 Experimental Structure

The experiment is organized into two stages to evaluate design choices and benchmark performance.

Controlled Variables

All experiments were run under the same conditions, including dataset, preprocessing, training setup, optimizer settings, evaluation metrics, and hardware. This ensures that performance differences come from architectural changes.

Stage A: Ablation Study

The first stage evaluates the effect of self-attention. Multiple variants were tested, differing only in the presence and placement of self-attention. A baseline without self-attention was used for comparison. Other components and loss functions were kept the same. Hyperparameters were adjusted per variant to allow each model to reach stable performance.

Stage B: Comparative Evaluation

The second stage compares the best variant from Stage A with two reference models representing different design approaches. All models used the same dataset splits, preprocessing, and computational limits to ensure a fair comparison.

3.2.3 Experimental Procedure

Each model and variant followed these steps:

1. Architectures were implemented with consistent initialization and fixed random seeds
2. Models were trained under the same hardware conditions, with repeated runs to reduce variability
3. Training loss, evaluation metrics, runtime, and memory usage were recorded, with checkpoints saved at fixed intervals
4. The best variant was selected based on validation metrics and perceptual quality

5. Final evaluation was performed on a held-out test set using quantitative metrics and visual inspection, with a subset used for subjective evaluation

3.2.4 Robustness and Control Measures

All models were trained under the same conditions. Dataset splits, preprocessing, and hyperparameters were fixed. Only the placement of self-attention was changed, without altering model depth or capacity.

3.2.5 Evaluation and Statistical Analysis

Metrics were computed on the test set and reported as mean and standard deviation across runs. Statistical tests were chosen based on data distribution. Subjective scores were analyzed using ordinal methods.

3.2.6 Scope and Design Limitations

The study is limited to a single dataset and fixed resolution, with restricted hyperparameter tuning due to compute constraints. This improves control and clarity but may limit how well results generalize.

3.3 Research Environment

Got it—keeping it straightforward, clear, and practical:

This section explains the setup, software, and conditions used for training and evaluation. All experiments were run on a virtualized GPU through a cloud provider, which is common in academic and small-scale research when dedicated hardware isn't available.

The system used an NVIDIA V100 GPU. Even though it's powerful hardware, running it in a virtual environment came with some limits. These included using only one GPU,

no support for distributed training, and some performance overhead that affected memory and I/O. Because of this, the model design and training approach had to be adjusted.

To keep the comparison fair, all experiments were done on the same virtual machine with the same settings. The next sections describe the hardware, software, virtualization setup, and how consistency was maintained across runs.

3.3.1 Hardware Configuration

All experiments were conducted on a virtual machine instance with GPU pass-through hosted on institutional cloud infrastructure.

Table 3.1. Research Environment Hardware and Software Configuration

Virtual Machine Host	
Instance Name	ICTCHPCoo-VM01
Virtualization Platform	QEMU/KVM
GPU Allocation	Single dedicated GPU via vGPU technology
Graphics Processing Unit	
Model	NVIDIA GRID V100D-32A (virtualized Tesla V100)
Architecture	Volta (Compute Capability 7.0)
VRAM	32 GB HBM2
CUDA Cores	5,120 (full V100 specification)
Tensor Cores	640
Memory Bandwidth	900 GB/s
Driver Version	553.62
CUDA Version	12.4
Virtual CPU	
Model	QEMU Virtual CPU version 2.5+
Virtual Cores	2 processors @ 2.19 GHz
Architecture	x64-based processor
System Memory	
RAM	32.0 GB
Storage	1 TB
Software Environment	
Operating System	Windows 11
Architecture	64-bit operating system, x64-based processor
Python Environment & Key Dependencies	
Python	3.10.1
NumPy	1.25.2
PIL	11.3.0
Deep Learning Framework	
PyTorch	2.5.0 + cu124
CUDA Toolkit	12.4
Build Tools and Compilation Environment	
IDE / Compiler Suite	Microsoft Visual Studio 2022
C++ Toolset	MSVC v143
CUDA Compiler	nvcc (CUDA Toolkit 12.4)
CUDA Architecture Support	Compute Capability 7.0 (V100)
Build System	MSBuild / Visual Studio Build Tools
Language Support	C++, CUDA C
Debugger	Visual Studio Debugger
Target Platform	Windows x64
Development and Deployment	
Code Editor	VS Code
Version Control	Git
Experiment Tracking	Local logging
Remote Access	AnyDesk

3.4 Flickr-Faces-HQ Dataset (FFHQ)

The Flickr-Faces-HQ (FFHQ) dataset was used for training and evaluation. It contains face images with variation in pose, expression, age, gender, and lighting.



Figure 3.2. FFHQ dataset samples.

FFHQ was selected because it is used in the original R3GAN framework, allowing consistent adaptation with prior work. Using faces reduces dataset variation compared to general super-resolution benchmarks.

Restricting the dataset simplifies analysis of self-attention and super-resolution effects. The consistent structure of faces also improves training stability under limited compute.

This choice supports the goal of analyzing architectural behavior rather than generalization across domains.

3.5 Data Preprocessing Steps

Data preprocessing was used to construct paired low-quality (LQ) and high-quality (HQ) images for supervised super-resolution training. All FFHQ images were resized to 256×256 and used as HQ targets.



Figure 3.3. Left: downsampled input. Right: 256×256 ground truth.

For each HQ image, an LQ version was generated by downscaling to 64×64 using the Lanczos algorithm, followed by additional degradations. Facial structure and alignment were preserved.

The dataset consists of paired samples, where each 64×64 image maps to a 256×256 target. This enables supervised learning from low- to high-resolution images. This setup isolates architectural effects, including self-attention placement, from data variation.

3.6 Model Architecture

This section describes the architecture of the models employed in the study, including the modified baseline R3GAN, the proposed self-attention-enhanced R3GAN variants used for ablation analysis, and the state-of-the-art models utilized for comparative

evaluation. All architectures were adapted or configured to support conditional fixed-scale single-image super-resolution from 64x64 to 256x256.

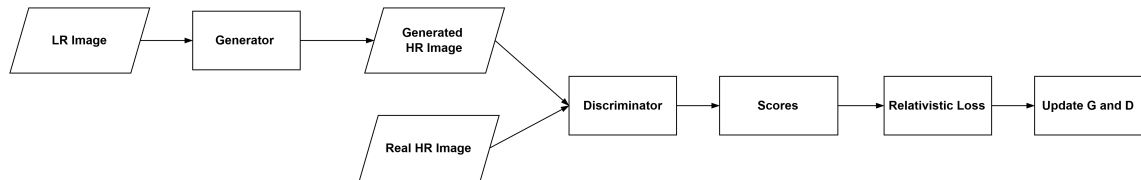


Figure 3.4. General overview of our modified R3GAN architecture process.

3.6.1 Baseline R3GAN Architecture

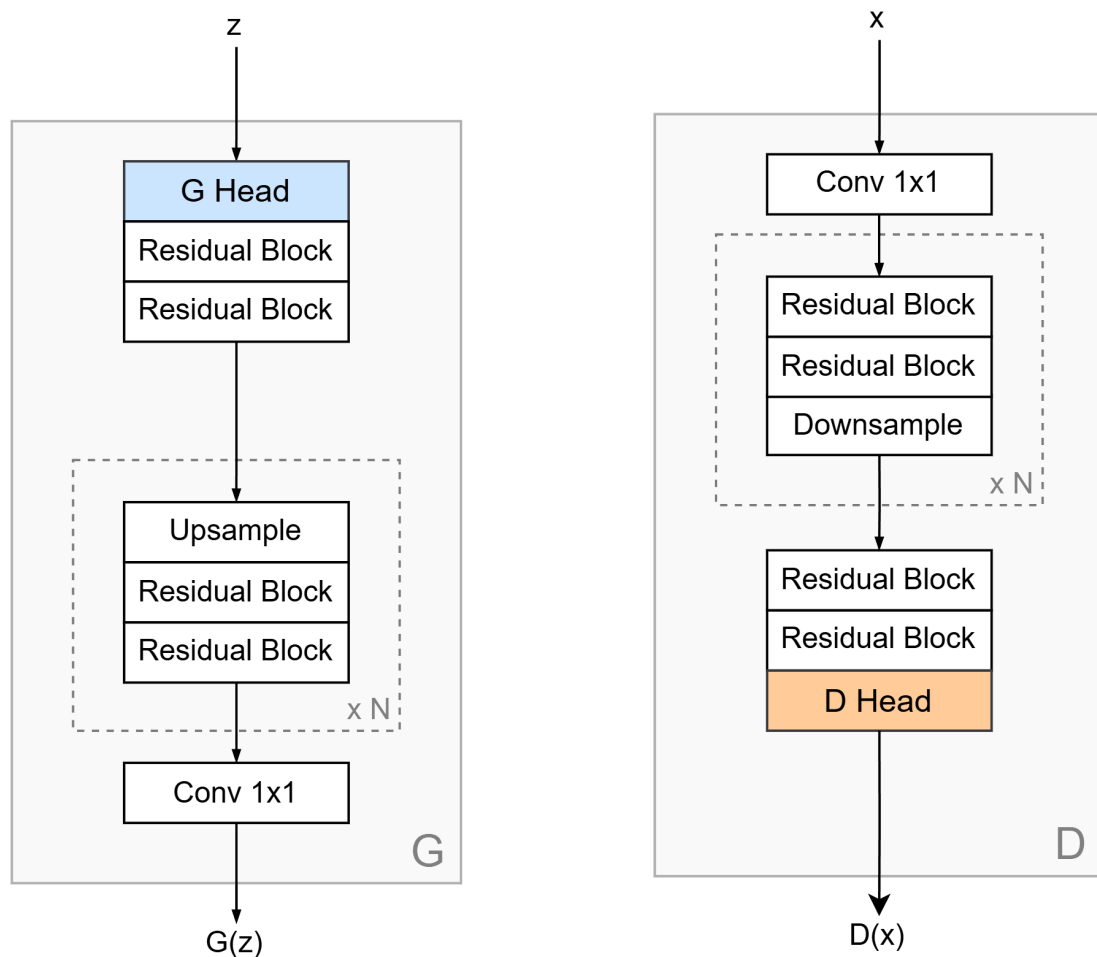


Figure 3.5. Illustration of the architecture for image generation using R3GAN.

The baseline model employed in this study was an adapted R3GAN architecture configured for conditional super resolution. Otherwise, the architecture is identical to the generative counterpart in prior literature. This adapted baseline served as the control model and did not include any self-attention mechanism.

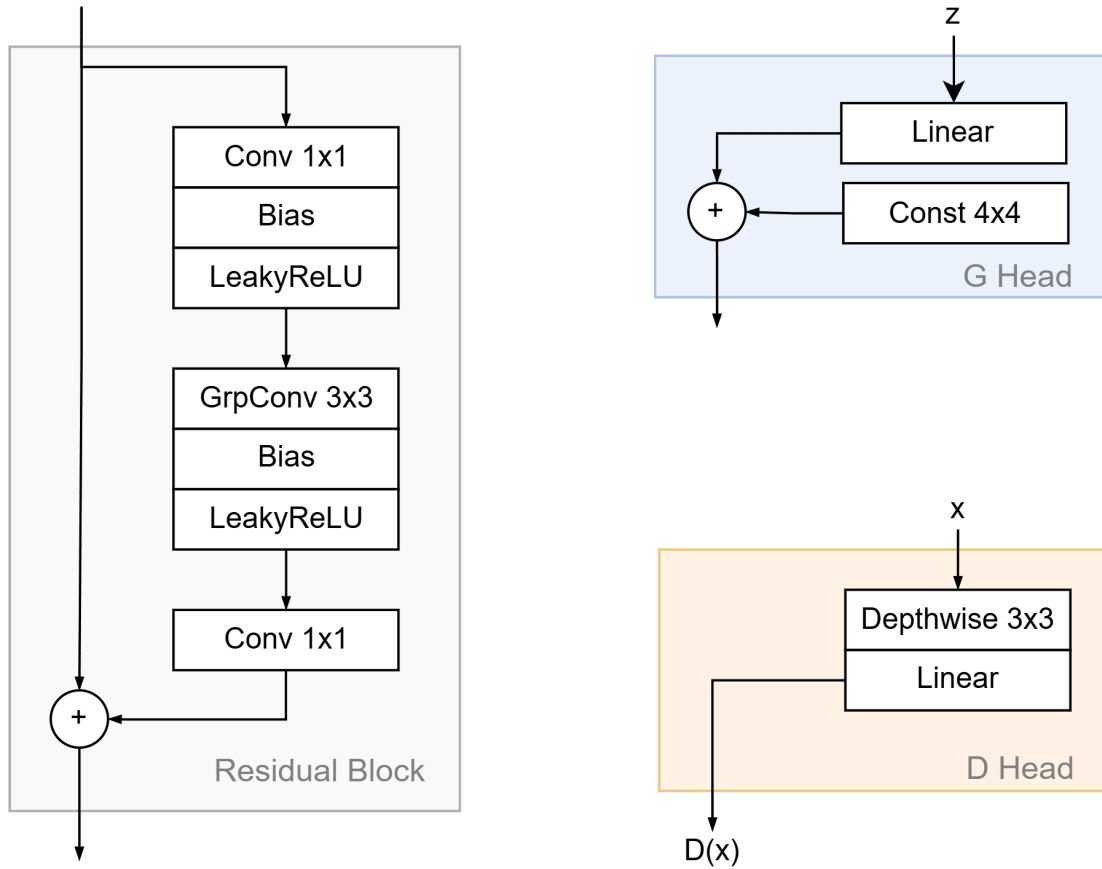


Figure 3.6. Structure of the Residual Block, Generator Head, and Discriminator Head.

The Residual Block serves as the primary feature transformation unit of R3GAN. It consists of a 1×1 convolution, a grouped 3×3 convolution, and another 1×1 convolution, each followed by LeakyReLU, with a residual skip connection. This structure supports gradient flow, channel mixing, and feature refinement without normalization layers, consistent with R3GAN design.

The Generator Head (G Head) starts generation by projecting the latent vector z through a linear layer and combining it with a learned constant feature map. In the

original R3GAN, this constant tensor is designed for unconditional image synthesis. For super-resolution, the G Head is modified to accept conditional inputs, allowing it to combine low-resolution image features with latent features before further processing. This shifts the generator's focus from image synthesis to high-frequency detail reconstruction.

The Discriminator Head (D Head) is also modified. In the original R3GAN, it uses a depthwise 4×4 convolution without padding followed by a linear layer, which quickly reduces spatial information. While suitable for unconditional generation, this is not ideal for super-resolution, where fine details are important.

To preserve spatial information, the 4×4 convolution is replaced with a padded depthwise 3×3 convolution applied before global average pooling. This modification preserves spatial resolution longer within the discriminator, allowing for gradual aggregation of local evidence across feature maps. Such behavior is crucial in super-resolution settings, as it enables the discriminator to evaluate texture consistency, edge sharpness, and local realism more effectively. Larger unpadded kernels were found to prematurely discard spatial information, leading to reduced texture diversity and weaker adversarial feedback.

3.6.2 Self Attention Module

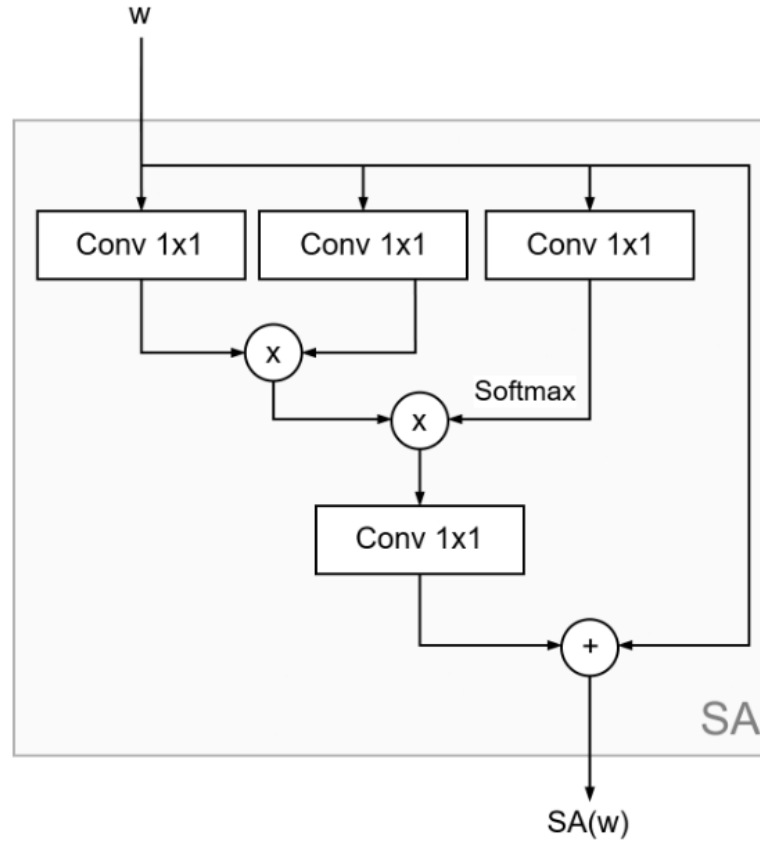


Figure 3.7. The concept of the non-local block integrated into the G and D.

In order to capture long-range dependencies in image or video features, we utilize the non-local operation introduced by Wang et al. (2018). Unlike traditional convolutional or recurrent operations that operate over local neighborhoods or sequential data, the non-local operation allows each output position to aggregate information from all positions in the input.

3.6.3 Self-Attention R3GAN Variants

Multiple self-attention-enhanced R3GAN variants were developed using a placement-based ablation strategy. The self-attention module described in Section 3.6.2 was integrated into the networks at selected intermediate stages, while all remaining architectural components were kept identical to the baseline R3GAN configuration to

focus only on the configuration of implementing self-attention on both networks rather than isolated placements.

Each variant differed in the location at which the self-attention layer was inserted within the networks, enabling systematic analysis of how attention placement influences feature representation and super-resolution performance. Placement choices follow the rationale introduced by Zhang et al. SAGAN literature, adopting this logic, guided the selection of candidate insertion points. This design ensured that the effects of self-attention could be isolated without introducing additional architectural or training-related variables

By maintaining consistent training settings, loss functions, and optimization strategies across all variants, the ablation study attributed observed performance differences solely to the inclusion and placement of the self-attention mechanism within the R3GAN architecture.

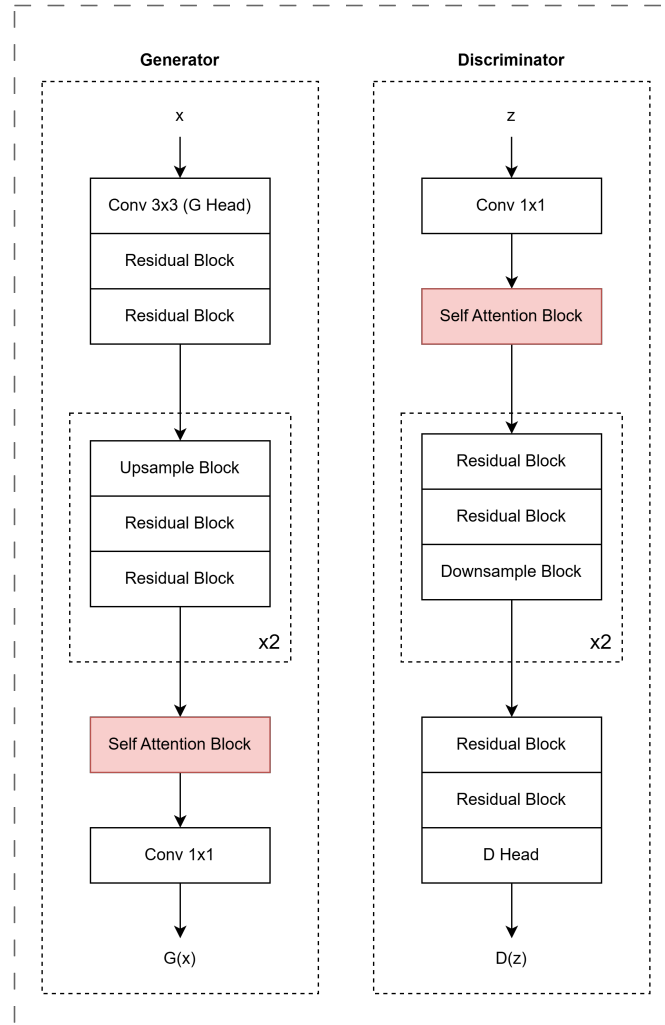


Figure 3.8. The architectural configuration of Self-Attention R3GAN Variant 1.

The placement of the self-attention modules in both networks followed a mirrored structural design following SAGAN's philosophy, ensuring balanced global feature modeling between the generator and discriminator.

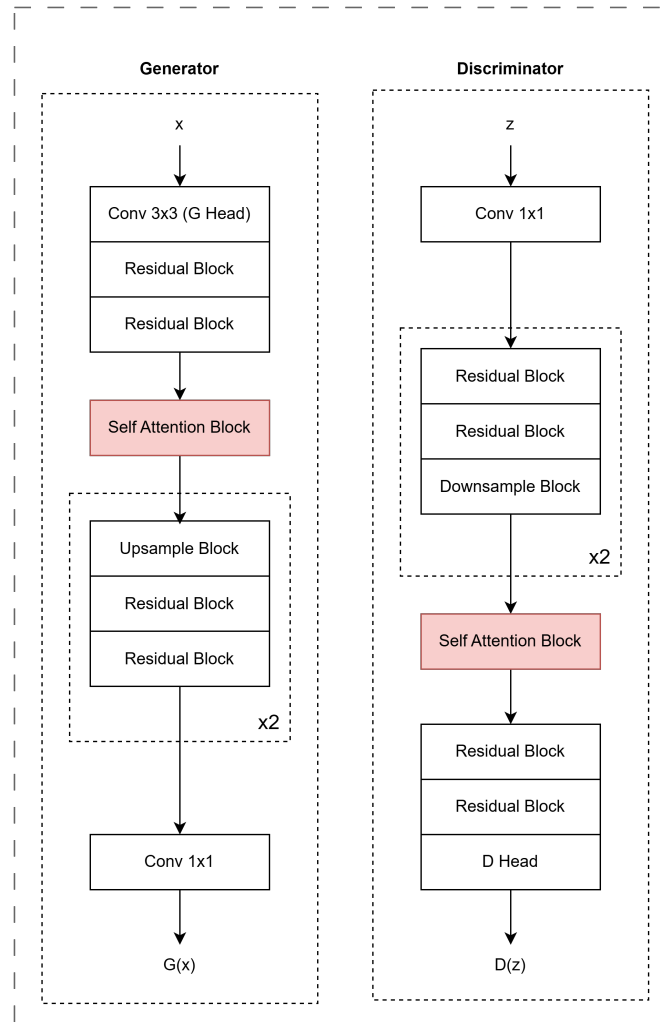


Figure 3.9. The architectural configuration of Self-Attention R3GAN Variant 2.

This placement strategy positions the self-attention mechanism at an earlier feature abstraction level in both networks, similar to Variant 1, with it being an inverted placement of the self-attention mechanism.

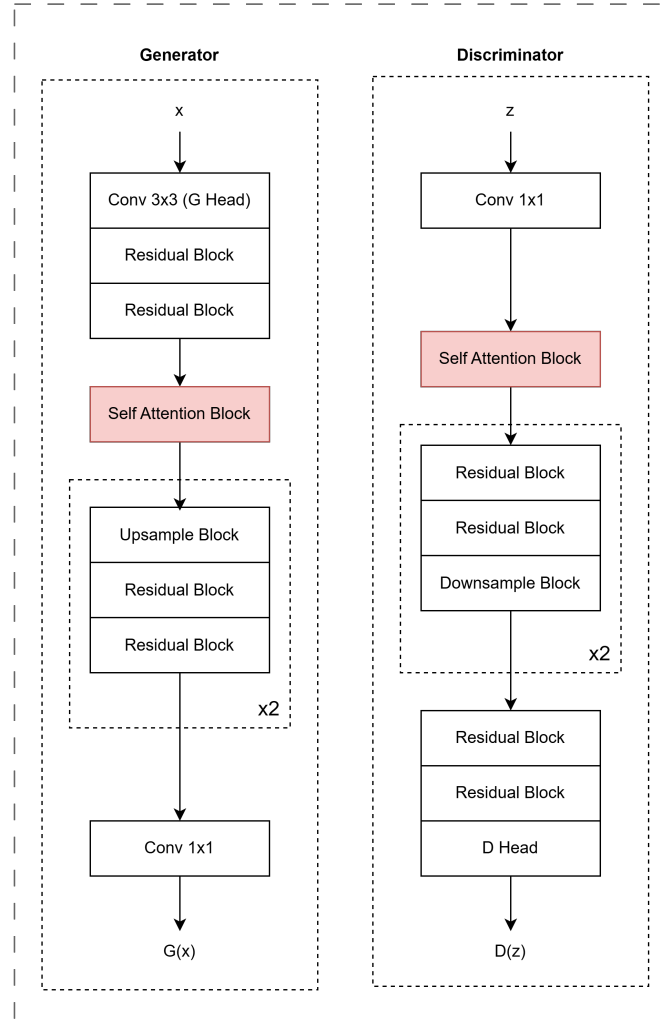


Figure 3.10. The architectural configuration of Self-Attention R3GAN variant 3.

This placement strategy positions the self-attention mechanism at an earlier feature abstraction level in both networks, enabling the modeling of global dependencies during the initial stages of feature transformation.

3.6.4 SwinIR and REAL-ESRGAN

This section presents the architectures and a brief overview of SwinIR and REAL-ESRGAN, which are included in the study to provide comparative context for evaluating the performance of the proposed Self-Attention R3GAN. These models were selected due to their established use in image enhancement and restoration literature and their

distinct architectural approaches.

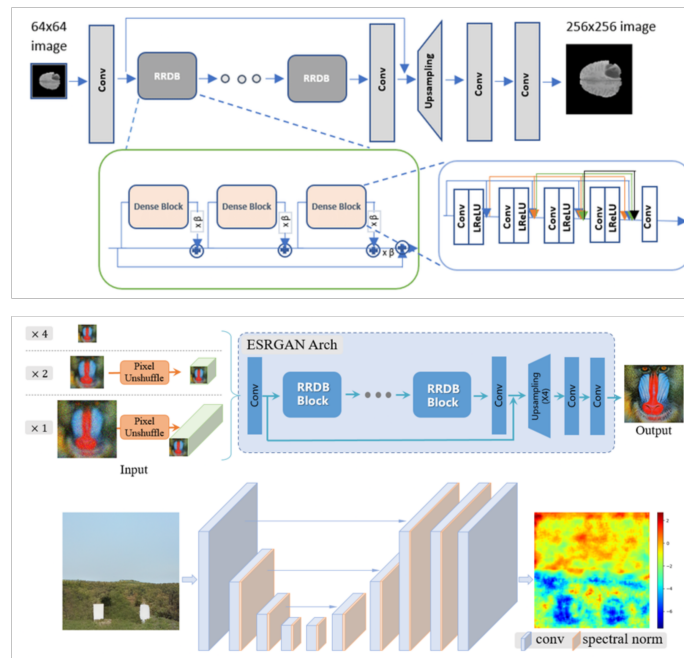


Figure 3.11. The architecture of REAL-ESRGAN (Wang et. al., 2021).

REAL-ESRGAN is a GAN-based image enhancement model that emphasizes perceptual quality through adversarial learning and the use of residual-in-residual dense blocks (RRDB). Its architecture is designed to enhance texture realism and high-frequency details, making it suitable for assessing the perceptual characteristics of the proposed GAN-based approach in comparison with an established convolutional adversarial framework.

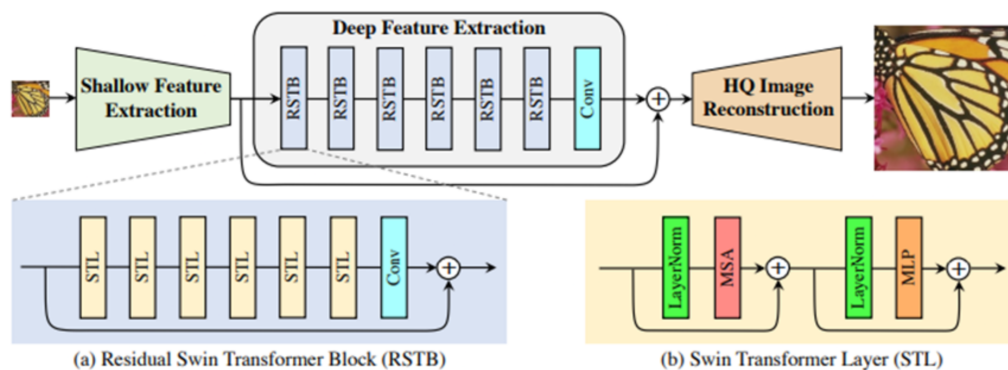


Figure 3.12. Architecture of SwinIR (Liang et. al., 2021)

SwinIR is a Transformer-based image restoration model that utilizes shifted window self-attention to capture long-range dependencies while maintaining computational efficiency. Unlike GAN-based models, SwinIR relies on attention-driven feature modeling without adversarial training. Its inclusion enables analysis of how the proposed self-attention-augmented GAN compares with a purely attention-based architecture for image enhancement tasks.

The inclusion of SwinIR and REAL-ESRGAN is intended to contextualize the performance of the proposed model rather than to establish state-of-the-art results. The comparative evaluation focuses on image quality, perceptual characteristics, and computational behavior under identical experimental conditions, providing insight into the strengths and limitations of the proposed architecture within a low-resource setting.

3.7 Training Strategy

This section describes the optimization process, loss functions, and training setup used for all models. The training strategy ensures stable adversarial learning, consistent comparison across architectures, and reproducibility under limited computational resources.

3.7.1 Overall Training Framework

The training strategy was designed for fair comparison across all R3GAN variants under limited computational resources. All models used the same training setup, with full use of available hardware and controlled training time. Differences between models were restricted to self-attention placement only, allowing performance changes to be attributed to this factor rather than training conditions.

3.7.2 Loss Function Formulation

The training objective consisted of three primary components:

1. Relativistic Adversarial Loss

The generator and discriminator were trained using a relativistic formulation where the discriminator evaluates whether real images are more realistic than generated images, rather than classifying them independently:

2. Generator Loss

$$\mathcal{L}_G = \mathbb{E}[\text{softplus}(-D(I_{LR}) - D(G(I_{HR})))] \quad \text{Equation (10)}$$

The generator loss is defined as $\mathcal{L}_G = \mathbb{E}_{I_{LR}}[\text{softplus}(-D(G(I_{LR})))]$, where $\mathbb{E}[\cdot]$ denotes the expectation over all samples in a training batch, $\text{softplus}(x) = \log(1 + e^x)$ is the softplus activation function, I_{LR} represents the low-resolution input image, and I_{HR} denotes the corresponding high-resolution ground truth. The generator output $G(I_{LR})$ refers to the super-resolved image produced from the low-resolution input, while $D(x)$ indicates the scalar output of the discriminator reflecting the perceived realism of an input image x . Accordingly, $D(G(I_{LR}))$ represents the discriminator's realism score for generated images, and $D(I_{HR})$ denotes the score assigned to real high-resolution images. Implementation of the generator loss is to make the discriminator think the generated images are more realistic than the actual ground truth image.

3. Discriminator Loss

$$\mathcal{L}_D^{\text{adv}} = \mathbb{E}[\text{softplus}(-D(I_{HR}) - D(G(I_{LR})))] \quad \text{Equation (11)}$$

The discriminator adversarial loss is defined as $\mathcal{L}_D^{\text{adv}} = \mathbb{E}_{I_{HR}}[\text{softplus}(-D(I_{HR}))] + \mathbb{E}_{I_{LR}}[\text{softplus}(D(G(I_{LR})))]$, where $\mathbb{E}[\cdot]$ denotes the expectation over all samples

in a training batch, $\text{softplus}(x) = \log(1 + e^x)$ is the softplus activation function, I_{LR} represents the low-resolution input image, and I_{HR} denotes the corresponding high-resolution ground truth. $D(I_{HR})$ is the discriminator's realism score for real images, while $D(G(I_{LR}))$ is the score for super-resolved images generated by the generator. The objective of the discriminator is to assign higher scores to real images than to generated ones, thereby improving its ability to distinguish between authentic and synthesized high-resolution samples. While the gradient penalty is defined as

$$\mathcal{L}_D = \mathcal{L}_D^{\text{adv}} + \frac{\gamma}{2}(R_1 + R_2) \quad \text{Equation (12)}$$

To further stabilize adversarial training, a gradient penalty term is incorporated into the discriminator loss. This penalty is weighted by a coefficient γ and consists of two components: an R_1 regularization term applied to real images, defined as

$$R_1 = \mathbb{E} \left[\|\nabla_{I_{HR}} D(I_{HR})\|_2^2 \right] \quad \text{Equation (13)}$$

and an R_2 regularization term applied to generated images, defined as

$$R_2 = \mathbb{E} \left[\|\nabla_{G(I_{LR})} D(G(I_{LR}))\|_2^2 \right] \quad \text{Equation (14)}$$

Here, ∇ denotes the gradient operator, and $\|\cdot\|_2^2$ represents the squared L_2 -norm. These gradient penalties constrain the magnitude of discriminator gradients, discouraging overly confident predictions and promoting smoother decision boundaries, which collectively contribute to improved training stability. The purpose of gradient penalties is to regularize the discriminator to prevent it from becoming too confident or unstable. They penalize large gradients, leading to smoother decision boundaries.

3.7.3 Optimization Configuration

Optimizer Settings

Both the generator and discriminator used the Adam optimizer with a learning rate of 1×10^{-4} and $\beta_1 = 0.0, \beta_2 = 0.99$. The use of $\beta_1 = 0.0$ follows GAN training practices (Hinz et al., 2020) to reduce oscillations during adversarial optimization.

Training settings were adjusted per variant based on stability and convergence. Standard runs used batch size 8 for 62,500 steps (50 epochs). Models that improved beyond typical convergence used smaller batch sizes and longer training to fully use their capacity. This allows each variant to reach its performance limit under similar computational constraints.

Learning Rate Scheduling

A cosine annealing schedule with linear warmup was used for both networks. During a 500-step warmup, the learning rate increased linearly from 0 to 1×10^{-4} . After warmup, it decayed following a cosine schedule to a minimum of 2×10^{-5} . This reduces early instability and supports convergence.

3.7.4 Exponential Moving Average (EMA)

An exponential moving average (EMA) of generator weights was maintained during training to stabilize inference:

$$\theta_{\text{EMA}} \leftarrow \beta_{\text{EMA}} \theta_{\text{EMA}} + (1 - \beta_{\text{EMA}}) \theta_G \quad \text{Equation (15)}$$

where θ_{EMA} is the EMA weights, θ_G is the generator weights, and β_{EMA} controls the smoothing factor. The EMA model was used exclusively for evaluation and final inference, while training updates were applied to the primary generator weights. This approach reduces high-frequency fluctuations in generated image quality and is particularly important for super-resolution, where small parameter fluctuations can

lead to noticeable texture instability.

3.7.5 Training Procedure

The training process followed an alternating optimization scheme between the discriminator and generator. For each iteration, the discriminator was updated by sampling a batch of lower-resolution and high-resolution image pairs (I_{LR}, I_{HR}) . The generator produced super-resolved images $I_{SR} = G(I_{LR})$, which were then used to compute the discriminator’s adversarial loss, including the applied gradient penalties, followed by parameter updates via backpropagation. Subsequently, the generator was updated using a newly sampled batch of (I_{LR}, I_{HR}) pairs, where the adversarial loss computed from the generated super-resolved outputs was used to update the generator parameters. After both updates, an exponential moving average (EMA) of the generator weights, θ_{EMA} , is updated to stabilize training. This sequence was repeated for every training iteration, with both networks updated once per batch.

3.7.6 Controlled Variant Training

To isolate the effects of self-attention placement, all R3GAN variants were trained to maximize hardware capabilities while maintaining largely similar configurations, with the exception of a single model due to computational overhead. The training setup included the same dataset and preprocessing pipeline, identical optimizer hyperparameter and learning rate schedule, a uniform training duration of 50 epochs, the same hardware configuration, and consistent gradient penalty coefficient ($\gamma = 10.0$) and EMA decay schedule. The same batch size was used for all variants, except Variant 2, which used a batch size of 4 due to memory limits caused by the self-attention module. Full architectural details are provided in Section 3.6.3.

3.7.7 Training Stability and Monitoring

Training stability was monitored through regular evaluation of generator adversarial loss (L_G), the discriminator adversarial loss ($L_{D_a} dv$), and the R_1 and R_2 gradient penalties on real and generated images, respectively, which were recorded every 500 iterations. Visual inspection was performed by saving generated super-resolution outputs at the same interval, including side-by-side comparisons of low-resolution inputs, super-resolved results, and high-resolution ground truth images. Quantitative evaluation relied primarily on the Fréchet Inception Distance (FID), computed every 1,000 iterations using 50,000 samples, which served as the main criterion for convergence and model selection. To ensure reproducibility and training continuity, model checkpoints were saved every 2,000 steps, with the best-performing checkpoint retained based on FID improvements and the most recent checkpoint preserved for training resumption. Other evaluation metrics (see Section 3.8) were evaluated post training on the final models rather than during the training process. This approach reduced computational overhead during training while still enabling a comprehensive quality assessment of the final models.

3.7.8 Implementation Details

The training pipeline was implemented in PyTorch 2.5.0+cu124, and mixed-precision training was not used, instead using the full FP32 training. Gradient accumulation was not applied, so the batch size directly corresponded to the effective batch size. No data augmentation techniques were applied, and gradient clipping was also omitted, as training stability was sufficiently ensured through gradient penalty terms. Model weights were initialized using the Mean Scale Rule (MSR), where weights were sampled from a normal distribution $W \sim \mathcal{N}\left(0, \frac{\text{gain}}{\sqrt{\text{fan_in} \times \text{kernel_size}^2}}\right)$. The gain parameter was adjusted according to the activation function, specifically LeakyReLU with $\alpha = 0.2$, $\text{gain} \approx \sqrt{2}/(1 + 0.2^2)$.

3.8 Evaluation metrics

To comprehensively assess the effectiveness of the proposed augmentation of the architecture and its conversion to a conditional single-image super-resolution task, both objective and subjective evaluation metrics were employed. Objective metrics quantify reconstruction fidelity and perceptual similarity, while subjective evaluation captures human visual preference. Performance was evaluated using PSNR, SSIM, FID, and subjective metrics including Mean Opinion Score (MOS) and blind ranking. These metrics measure pixel accuracy, perceptual quality, distribution similarity, and human perception.

3.8.1 Peak Signal-to-Noise Ratio (PSNR)

PSNR measures pixel-level reconstruction accuracy between generated super-resolved images and ground truth. It is based on the ratio between the maximum pixel value and the mean squared error (MSE). Higher values indicate lower distortion.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right) \quad \text{Equation (16)}$$

MAX_I is the maximum possible pixel value, and MSE is the mean squared error between generated and reference images.

3.8.2 Structural Similarity Index Measure (SSIM)

SSIM measures perceptual similarity between generated and ground-truth images by comparing luminance, contrast, and structure. Compared to PSNR, it better reflects human visual perception and structural consistency.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad \text{Equation (17)}$$

SSIM ranges from -1 to 1 , where values closer to 1 indicate higher similarity and

better structural preservation.

3.8.3 Fréchet Inception Distance (FID)

FID measures distribution similarity between generated super-resolved images and real high-resolution images. Unlike PSNR and SSIM, it compares feature distributions extracted from a pretrained Inception network.

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right) \quad \text{Equation (18)}$$

Lower FID indicates closer alignment between real and generated distributions, reflecting better realism and diversity. In this study, FID is the main metric for comparing model variants.

3.8.4 Survey-Based Mean Opinion Score (MOS)

MOS was used to measure perceived image quality. 70 participants rated images on a 1–5 scale based on realism, clarity, and overall quality. Images were shown in random order, with a short calibration phase for consistency.

MOS is computed as:

$$\text{MOS} = \frac{1}{N} \sum_{n=1}^N R_n \quad \text{Equation (19)}$$

where R_n is the score for each image.

3.8.5 Blind Subjective Ranking Evaluation

A blind ranking test was also performed. For each image, outputs from different models were shown in random order. Participants assigned ranks $R_{i,j}$, where 1 is best and higher values indicate lower quality. The average rank per model is $\bar{R}_i$.

Statistical significance was tested using the Friedman test:

$$\chi_F^2 = \frac{12}{Nk(k+1)} \sum_{i=1}^k \bar{R}_i^2 - 3N(k+1) \quad \text{Equation (20)}$$

where N is the number of images, k is the number of models, and $\bar{R}_i$ is the mean rank.

If significant differences were found, pairwise comparisons used the Wilcoxon signed-rank test:

$$W^+ = \sum_{d_i > 0} \text{rank}(|d_i|), \quad W^- = \sum_{d_i < 0} \text{rank}(|d_i|), \quad W = \min(W^+, W^-) \quad \text{Equation (21)}$$

where d_i is the difference between paired ranks.

CHAPTER 4. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results of adding a self-attention mechanism (non-local block) to the R3GAN architecture for fixed-scale single-image super-resolution. The variants are first compared through an ablation study. The selected variant is then compared with state-of-the-art super-resolution models using performance, stability, and perceptual quality.

4.2 Evaluation Context

The evaluation looks at model behavior during and after training. Hardware utilization is included because the training environment affected the experiments. The Super-Resolution Self-Attention R3GAN (SR-SAR3GAN) variants are compared to choose the configuration used for comparative analysis and subjective evaluation.

4.3 Hardware Utilization During Training

During the ablation experiments, GPU utilization peaked at approximately 100%, while CPU utilization averaged around 80%. RAM utilization reached 31GB for batch size 8 training. For batch size 4, RAM use averaged around 16-18GB. The workload was mainly GPU-bound because of the model architecture and training procedure.

4.4 Ablation Study of Self-Attention Placement

The main hypothesis of this study is that R3GAN can be used as a baseline for single-image super-resolution, and that its performance changes when self-attention is placed in different parts of the network.

The discussion starts with the R3GAN-based super-resolution model and the baseline version without self-attention. The baseline is used as the reference model for checking how much the self-attention module contributes to the results.

4.4.1 Baseline R3GAN as SR



Figure 4.1. G and D loss over the training steps.

The generator loss stays around 0.74-0.76 for most of the training. The discriminator loss is less stable at the start, ranging from 0.66 to 0.84, but later settles around 0.64-0.67. For most steps, the discriminator loss is slightly lower than the generator loss. The usual Discriminator-to-Generator ratio is around 0.87 to 0.90.

The discriminator does not overpower the generator in this setup. If it did, the

generator could receive weak gradients and the training could collapse. At the same time, the discriminator is still able to guide the generator. Since both losses settle instead of diverging, the training reaches a balanced adversarial state.

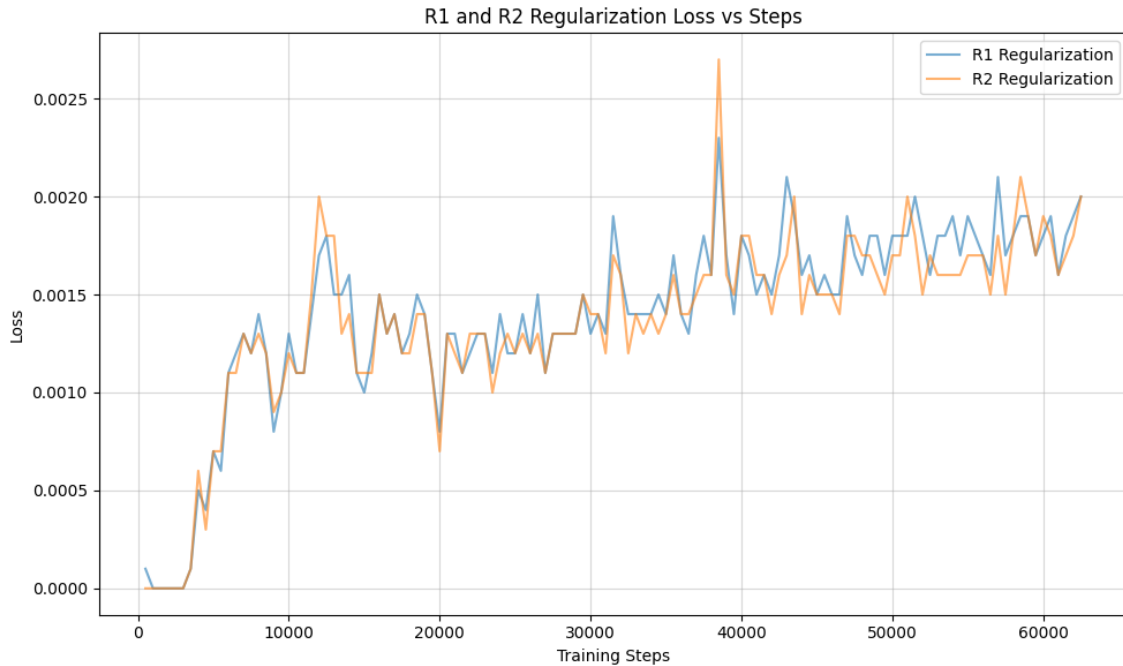


Figure 4.2. R1 and R2 gradient regularization values over 65,000 training steps.

As shown in Figure 4.2, R1 and R2 remain near 0.0000-0.0001 during the first 3,000 steps. These terms penalize the squared norm of discriminator gradients with respect to the input images. In early training, the discriminator gradients are still noisy and not well organized, so the penalty values remain low.

R1 and R2 increase at steps 4,000 and 10,000, reaching around 0.0011-0.0013. From step 20,000 onward, they settle around 0.0016-0.0020, with occasional spikes up to 0.0027. At this point, the discriminator is working near its limit while R1 and R2 keep it from becoming too aggressive.

The higher gradient penalties in late training point to a possible capacity limit. The discriminator is still trying to separate real and generated images, but the regularization terms are already constraining it. Under this setup, further improvement would likely need architectural changes, not only more training time.

4.4.2 Super-Resolution Self-Attention R3GAN

All SR-SAR3GAN variants include a self-attention module. The main difference is where the module is placed in the generator and discriminator, as described in Chapter 3. Because of this, the comparison focuses on attention placement.

SR-SAR3GAN-1

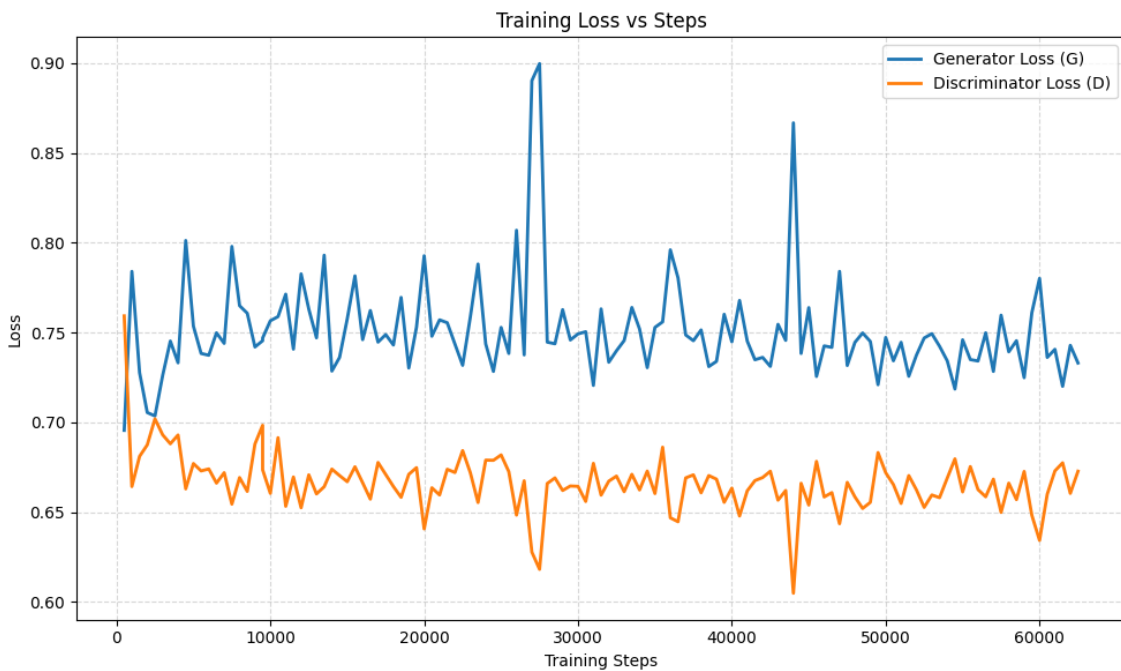


Figure 4.3. G and D loss of SR-SAR3GAN-1.

Around steps 27,000–27,500, the generator loss rises to 0.89–0.90, while the discriminator loss drops to 0.62. The change marks a short disturbance in the adversarial balance. A likely reason is that the generator improves its outputs in a way that the discriminator has not yet adapted to.

These spikes do not show training collapse. They are better treated as transition points where the generator learns patterns that affect discriminator performance.

The discriminator loss ranges from 0.60 to 0.69, which is wider than the baseline range of 0.64 to 0.67. Drops at steps 27,500 (0.618) and 44,000 (0.605) happen near

generator loss spikes in Figure 4.3. In this pattern, generator changes are followed by discriminator adjustment.

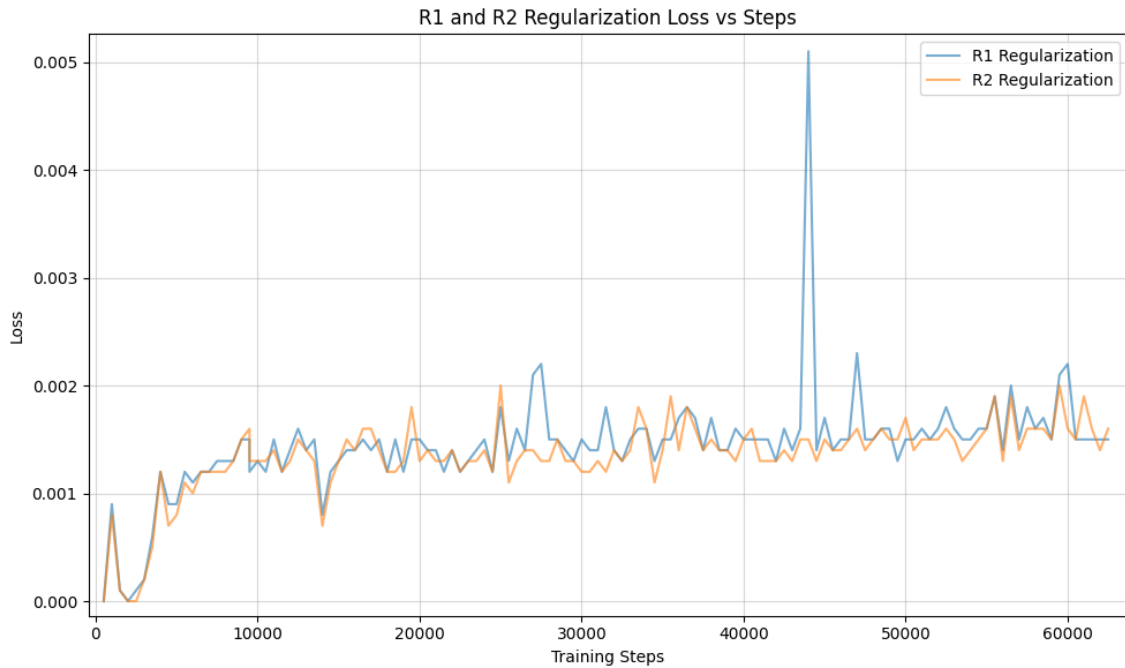


Figure 4.4. R1 and R2 regularization of SR-SAR3GAN-1.

Compared with the baseline, R1 and R2 activate earlier in SR-SAR3GAN-1. They reach 0.0009-0.0012 by step 1,000, while the baseline remains near zero until step 4,000. From steps 5,000-25,000, the values stay around 0.0011-0.0018.

From steps 25,000-65,000, R1 and R2 usually stay around 0.0014-0.0017, with spikes to 0.0020-0.0023. At step 44,000, R1 reaches 0.0051, about 3x the usual value. The spike happens when the generator loss reaches 0.867, meaning the discriminator gradients became steep while separating the generated images from the ground truth.

SR-SAR3GAN-2

Variant 2 places attention in the early stage of the generator and the later stage of the discriminator. Training this setup with batch size 8 caused frequent crashes, so batch size 4 was used. To keep the sample exposure comparable, the number of training steps was doubled. All models still process an equivalent number of samples.

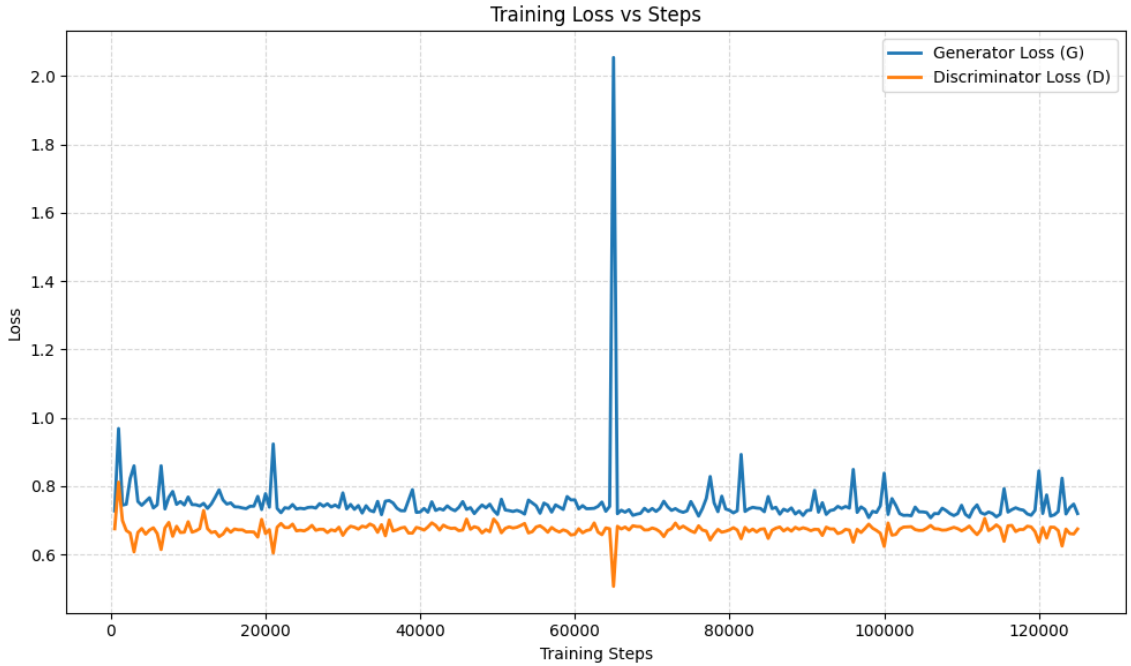


Figure 4.5. G and D loss of SR-SAR3GAN-2.

The generator loss in Variant 2 stays mostly between 0.71 and 0.76, with some spikes. It is less rigid than the baseline but less volatile than Variant 1.

Spikes occur at steps 1,000 ($G=0.969$), 3,000 ($G=0.85$), 6,500 ($G=0.86$), 21,000 ($G=0.92$), and later at steps 65,000 ($G=2.05$), 77,500 ($G=0.82$), 81,500 ($G=0.89$), 96,000 ($G=0.84$), 100,000 ($G=0.83$), 120,000 ($G=0.84$), and 123,000 ($G=0.82$). The largest spike is at step 65,000, where the generator reaches 2.055 and the discriminator loss drops to 0.507. After about 500 steps, the values return to $G=0.719$ and $D=0.683$, so the model recovers instead of collapsing.

The discriminator loss ranges from 0.60 to 0.70. It drops at steps 3,000 (0.608), 6,500 (0.614), 21,000 (0.604), and 65,000 (0.507). These drops match the generator spikes and FID improvements, so they mark training transitions.

The D/G ratio stays around 0.88–0.92. When the generator improves, the discriminator reacts afterward. The result is a step-by-step pattern where generator progress forces the discriminator to adjust.

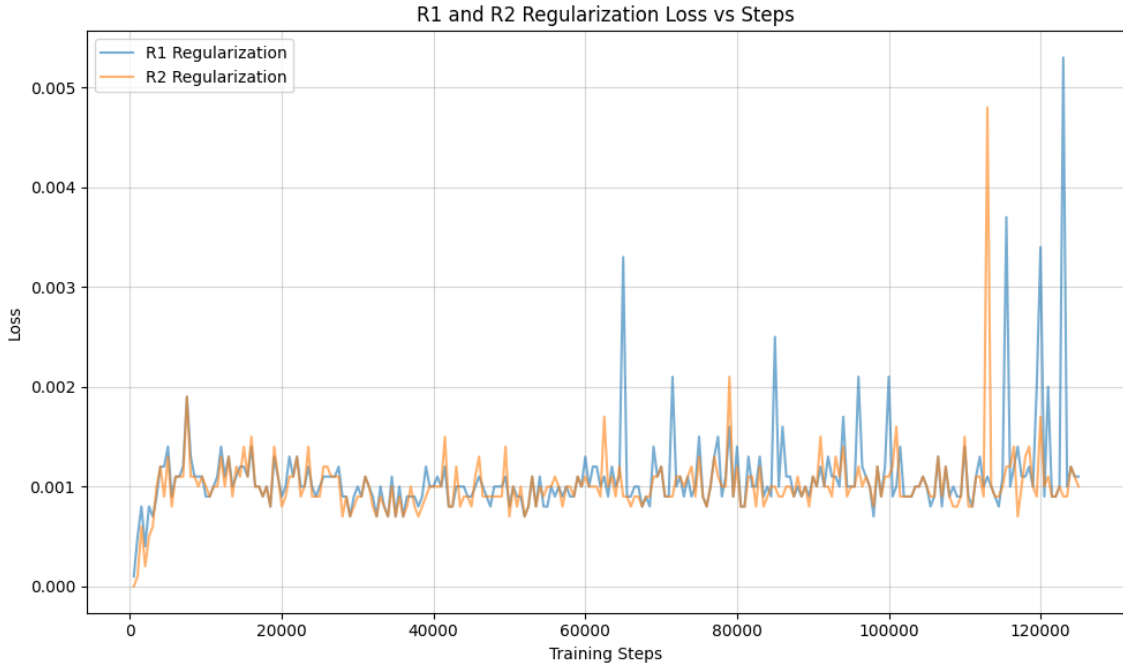


Figure 4.6. R1 and R2 regularization of SR-SAR₃GAN-2.

Variant 2 has moderate gradient penalties in early training. R1/R2 starts around 0.0001–0.0005 at step 500 and rises to 0.0012–0.0013 by step 5,000. The gradient pattern remains stable, without the early instability seen in input-level attention variants.

During steps 5,000–40,000, the penalties stay within 0.0007–0.0015. Most values are around 0.0009–0.0011, which is lower and less variable than Variant 1 and the baseline.

In late training, from steps 40,000–125,000, penalty spikes align with generator improvements. At step 65,000, R1 rises to 0.0033 while R2 stays at 0.0009, giving a 3.7× difference. The gap reflects sharper discriminator gradients while the model adapts to better generator outputs. R3 regularization reduces these updates.

Other spikes occur at steps 71,500 (R1=0.0021), 85,000 (R1=0.0025, R2=0.0010), 113,000 (R2=0.0048), 115,500 (R1=0.0037), 120,000 (R1=0.0034, R2=0.0017), and 123,000 (R1=0.0053). These events match the late-stage generator spikes, while the other variants have already plateaued. Variant 2 continues learning later in training.

SR-SAR₃GAN-3

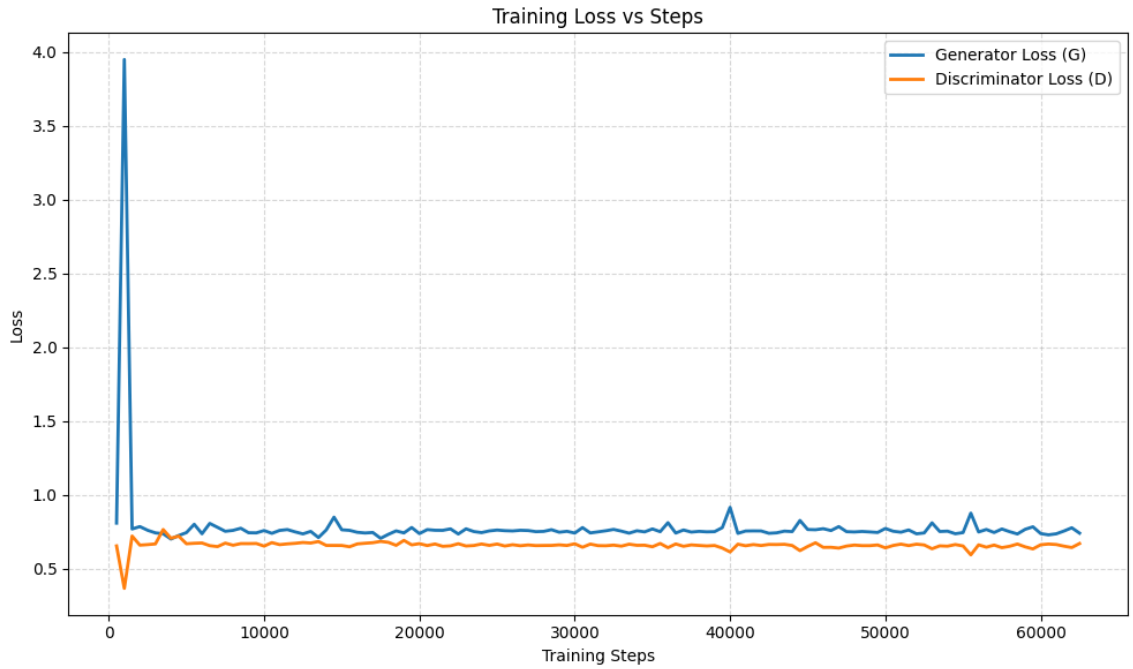


Figure 4.7. G and D loss of SR-SAR₃GAN-3.

Variant 3 has an early G loss spike to 3.950. The value is more than 5x the usual equilibrium level and almost double Variant 2's largest spike of 2.055. At the same time, D loss drops to 0.367, meaning the discriminator can separate the generated outputs from the ground truth easily.

The result is linked to placing self-attention too early in the architecture. At that point, the features are still weakly processed, so attention can amplify noise. By step 1,500, the model recovers to $G=0.769$ and $D=0.721$.

After recovery, G loss usually stays within 0.73-0.78. It has moderate spikes at step 4,500 ($G=0.806$), step 6,500 ($G=0.807$), step 14,500 ($G=0.849$), step 30,500 ($G=0.779$), step 36,000 ($G=0.812$), step 40,000 ($G=0.915$), step 44,500 ($G=0.827$), step 53,000 ($G=0.811$), step 55,500 ($G=0.876$), and step 59,500 ($G=0.784$). Compared with Variant 2, the spikes are smaller after recovery. Variant 3 therefore explores more cautiously after the early instability.

The discriminator loss becomes rigid after the early disturbance. At step 1,000, $D=0.367$, showing that the discriminator dominates the generator. By step 1,500, D increases to 0.721, which is above the usual equilibrium range. From step 3,500 onward, D loss stays around 0.64-0.68. The D/G ratio stabilizes around 0.86-0.89, but the low variance suggests less useful adaptation than Variant 1 or Variant 2.

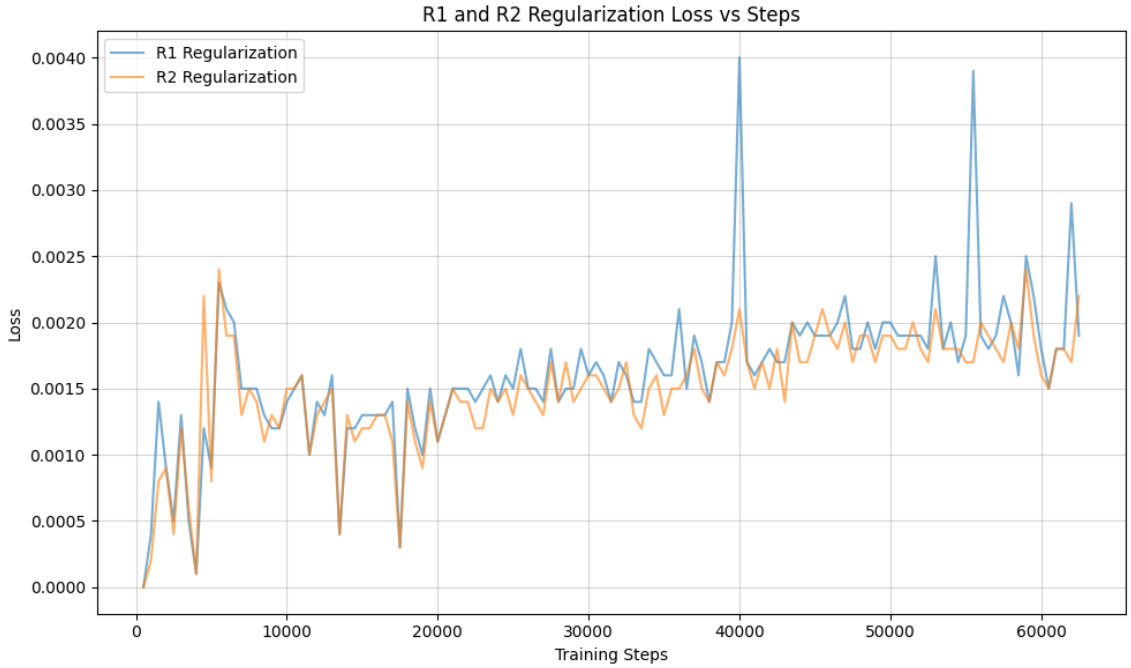


Figure 4.8. R1 and R2 regularization of SR-SAR₃GAN-3.

Variant 3 has the highest regularization levels among the variants. The baseline stays around 0.0016-0.0020, Variant 1 around 0.0014-0.0017, and Variant 2 around 0.0007-0.0012. Variant 3 stays around 0.0014-0.0020 for most of training, with increases to 0.0022-0.0025.

High points include step 40,000 ($R1=0.0040$, $R2=0.0021$) and step 53,000 ($R1=0.0025$, $R2=0.0021$). Other high points occur at step 55,500 ($R1=0.0039$, $R2=0.0017$), step 59,000 ($R1=0.0025$, $R2=0.0024$), and step 62,000 ($R1=0.0029$, $R2=0.0017$). The discriminator is working harder than in the other variants, but the final quality is still lower. This points to an architectural limitation rather than a lack of training effort.

4.4.3 Impact of Self-Attention Placement

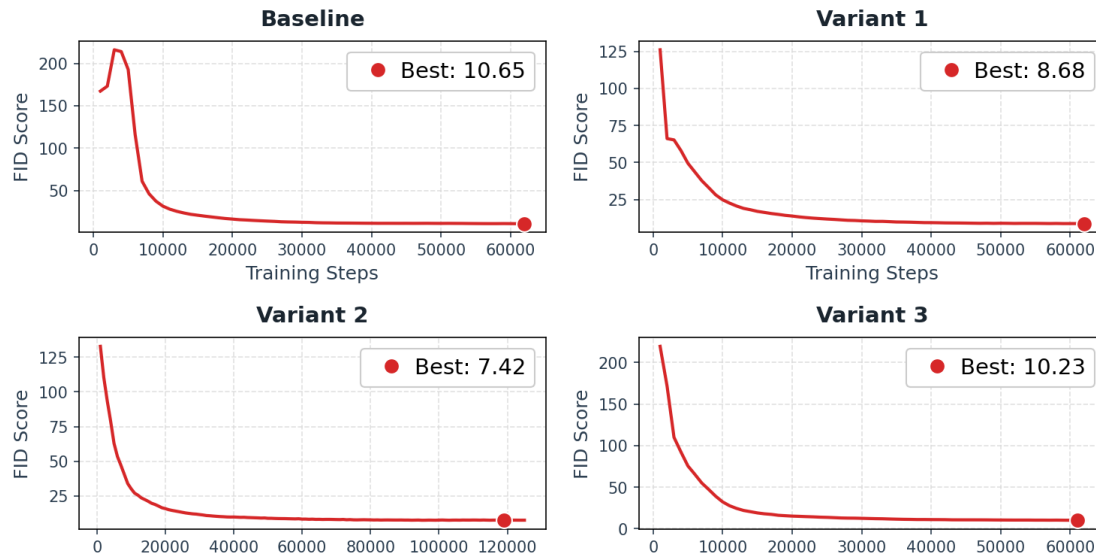


Figure 4.9. The FID over training steps for each model variant and the baseline.

Based on the figures in 1.9, The FID curve shows that the attention placement affects the early training of the models, The baseline starts at FID 167.23, rises to 216.05 at step 3,000, and decreases to 192.75 at step 5,000. Comparing the variants to the baseline, variant 1 starts at an FID score of 126.07 and declines without an early degradation phase compared to the baseline and was able to reach a 62% reduction, Its loss values fluctuate (G: 0.70–0.84, D: 0.64–0.70), matching the earlier observation that this variant explores feature representations during training. Variant 2 on the other hand starts at FID 132.85 and improves to 62.76 by step 5,000 with a 53% reduction, Its early convergence is slower than Variant 1, but it is steadier. The mid-level attention takes more iterations before the features stabilize. In later stages, it keeps stable values (G: 0.73–0.86, D: 10 0.61–0.81).

Lastly, Variant 3 starts with the weakest early behavior compared to the other variants with FID of 219.44, Its generator loss spike to 3.950 and discriminator drop to 0.367 show early instability. Gradient penalties help it recover, and it reaches FID 75.57 by step 5,000, but it remains behind the other attention variants relative to steps.

After step 5,000, all models move toward an FID range of around 14–16. By step 40,000, the models separate: Variant 1 (9.40), Variant 2 (9.74), Baseline (11.19), and Variant 3 (11.04). The attention-based models are lower than the baseline at this point.

From step 50,000 onward, the final ranking becomes stable: Variant 2 (7.42) > Variant 1 (8.68) > Baseline (10.65) > Variant 3 (10.23). Variant 2 achieves about 30% better FID than the baseline and about 15% better FID than Variant 1. Variant 1 improves over the baseline by 18.5%, while Variant 3 improves by only 3.9%. Attention placement matters, not only the presence of attention.

4.5 Selection of the Optimal R3GAN Variant

4.5.1 Summary of Model Configuration

Table 4.1. Training summary of all SR-SAR3GAN variants.

Hyperparameters	SR-R3GAN	SR-SAR3GAN-1	SR-SAR3GAN-2	SR-SAR3GAN-3
Resolution	$64 \times 64 \rightarrow 256 \times 256$	$64 \times 64 \rightarrow 256 \times 256$	$64 \times 64 \rightarrow 256 \times 256$	$64 \times 64 \rightarrow 256 \times 256$
Steps	62.5K	62.5K	125K	62.5K
Batch Size	8	8	4	8
Training Duration	8 Days	15 Days	11 Days	11 Days
Learning Rate	$1e-4$	$1e-4$	$1e-4$	$1e-4$
γ for R1 & R2	10	10	10	10
Adam β_2	$0.9 \rightarrow 0.99$	$0.9 \rightarrow 0.99$	$0.9 \rightarrow 0.99$	$0.9 \rightarrow 0.99$
EMA half-life (Mimg)	0.5	0.5	0.5	0.5
Channels per resolution	$64 \rightarrow 128 \rightarrow 256$	$64 \rightarrow 128 \rightarrow 256$	$64 \rightarrow 128 \rightarrow 256$	$64 \rightarrow 128 \rightarrow 256$
ResBlocks per resolution	2 - 2 - 2	2 - 2 - 2	2 - 2 - 2	2 - 2 - 2
Groups per resolution	4 - 4 - 4	4 - 4 - 4	4 - 4 - 4	4 - 4 - 4
G params	2.39M	2.39M	2.31M	2.39M
D params	2.39M	2.39M	2.47M	2.39M
Dataset	FFHQ	FFHQ	FFHQ	FFHQ

Batch Size Strategy

The baseline, Variant 1, and Variant 3 use batch size 8. It was the largest batch size that could fit within the GPU memory limit for 256×256 outputs and models with about 2.39M–2.40M parameters. Each update uses 8 LR–HR pairs, reducing gradient variance while giving 7,812 optimizer steps over 62,500 iterations.

Compared with a single-sample update, the variance reduction is about $\sqrt{8} \approx 2.8\times$. The setup gives a balance between stable gradients and enough update frequency.

Variant 2 uses a batch size of 4. The smaller batch makes the updates noisier and results in more updates per epoch. It also shows higher variation (G variance $\pm 0.03-0.05$), which leads to more trial-and-error behavior during training. For example, there is a spike at step 65,000 ($G = 2.055$).

The added noise appears to help in the later stages of training, as it allows the model to move beyond earlier learned patterns. The improvements of FID score from 10 to 7.4 occurs during this phase of the training.

Contrasting this, a batch size of 8 results in smoother and more stable training, but with less exploration. This is sufficient for simpler cases such as the baseline (G: 0.74 ± 0.02 , D: 0.65 ± 0.02). Variant 3 shows higher gradient penalty values ($R1/R2 = 0.0017-0.0029$), which suggests that the discriminator has more difficulty in adapting when the learning signal becomes noisier.

Dataset and Domain Specificity

The usage of the modified FFHQ dataset contains a 10,000 face images with variation in identity, expression, pose, and lighting. Selection of the dataset was necessary after earlier test with a general datasets produced inconsistent results and visible artifacts in its output, this change in dataset is based on the supervisor's recommendation. Images were downsampled from 256x256 ground truth images to 64x64 for training inputs using Lanczos filter, This defines a 4 $\times$ super-resolution task focused on faces, Since the dataset is face-specific, both the model and evaluation focus on facial structure and visual quality.

With a batch size of 8, each model processes 500,000 images over 62,500 steps, which is 12 about 7.1 passes through the dataset, Variant 2 uses a batch size of 4 but processes the same number of images over 125,000 steps.

The sampling pattern also differs by batch size, Batch size 8 produces 7,812 unique batches of 8 images, with more repeated combinations while batch size 4 produces 31,250 unique batches, allowing more varied pairings across the dataset.

Training Infrastructure and Reproducibility Considerations

The configuration table does not capture all the factors that affected training, Total training time varies across all the training runs: 8 days for the baseline, 15 days for Variant 1, and 11 days for Variants 2–3, even though the models have similar parameter counts and FLOPs. The longer runtime of Variant 1 (15 vs 8 days) is not due to model complexity alone, It is mostly caused by interruptions, storage delays, and shared GPU use.

Variant 1 had frequent restarts early in training (around steps 6,000–9,000), with interruptions about every 1,000 steps, Each restart requires reloading the model for training, resetting the CUDA kernels, restarting the data loader, and rebuilding the training state. With about 40–50% of steps affected, these overheads add several hours.

The remaining time difference is likely due to a resource sharing and hardware variations. Variant 2 runs at about 8.4 seconds per step, compared to 12.3 seconds for the baseline, even with similar compute cost. This points to differences in GPU usage, which is also reflected in the smoother training curve of Variant 2.

For this reason, training duration should not be treated as a fixed architectural cost. It is better understood as an upper bound affected by model design and the training environment.

Table 4.2. Quality metrics for the performance across all four model variants.

Model	↓ FID	SSIM ↑	PSNR ↑
BASELINE	10.65	0.69	23.98
SR-SAR3GAN-1	8.68	0.71	24.63
SR-SAR3GAN-2	7.42	0.72	24.85
SR-SAR3GAN-3	10.23	0.71	24.37

4.5.2 Quantitative Evaluation Results among SR-SAR3GAN

Quality Metrics Table

The metrics place SR-SAR3GAN-2 first among the four models. It has the lowest FID (7.42), along with SSIM 0.716 (+3% vs baseline) and PSNR 24.85 dB (+3.5% vs baseline). It is the only variant that performs well across all three metrics.

SR-SAR3GAN-1 ranks second in perceptual quality, with FID 8.68 (+18.5% vs baseline), SSIM 0.711 (+2.5%), and PSNR 24.63 dB (+2.6%). The baseline has the weakest FID (10.65) and lowest PSNR (23.98 dB), so it falls behind the attention-based models. SR-SAR3GAN-3 gives only a small improvement over the baseline, with FID 10.23, SSIM 0.708, and PSNR 24.37 dB. Early attention gives limited benefit despite adding complexity.

SSIM Performance analysis



Figure 4.10. Sample outputs of all developed models compared with the ground truth.

The SSIM scores are close across all models, ranging from 0.69–0.72. SR-SAR3GAN-2 achieves SSIM 0.72, which points to better structural preservation with mid-level attention.

SR-SAR3GAN-1 reaches an SSIM of 0.711, only about 0.7% lower than Variant 2, even though its FID is 17% worse (8.68 vs 7.42). The results shows that the SSIM does not separate the variants as clearly as FID. SSIM is mainly used to captures local details such as edges, texture, and contrast, while the FID metric reflects the overall consistency and how well the resulting output matches the ground truth images.

For SR-SAR3GAN-3 variant, it got an SSIM score of 0.708, which is slightly higher than the baseline score of 0.69. The improvements is miniscule, which is suggests that the placement of attention early has limited effect before the features are refined.

The baseline scoring 0.69 indicates that the convolutional model still preserves some of the local details, Nonetheless, its FID of 10.65 signify a weaker overall realism meaning that baseline keeps local structure but lacks the needed global consistency.

Peak Signal-to-Noise Ratio (PSNR): Pixel-wise Accuracy



Figure 4.11. The output of SR-SAR3GAN-2 (left) together with the ground truth (right).

SR-SAR3GAN-2 had reached a PSNR of 24.85 dB, corresponding to an average MSE of about 214 compared to 259 for the baseline which is around 21% lower. This results means a better pixel accuracy although GAN-based super-resolution still involves a trade-off between exact reconstruction and visual quality.

Its lower FID of 7.42 implies a better visual realism with more consistent textures specifically facial details even if some pixel differences remain.

SR-SAR3GAN-1 has a similar PSNR to SR-SAR3GAN-2 of 24.63 dB but a higher FID of 8.68, while its pixel accuracy is close to Variant 2, its overall visual quality and consistency is noticeably weaker.

Lastly, SR-SAR3GAN-3 reaches a PSNR score of 24.37 dB, only slightly above the baseline (+0.39 dB, about 10% lower MSE), Its FID improvement is small implying that early attention helps more with local details than the overall structure.

4.5.3 Metric Trade-off Analysis and Decision Criteria

The overall ranking is consistent across all of the variants, SR-SAR3GAN-2 performs the best among all of the variants, with FID of 7.42, SSIM 0.716, and PSNR 24.85 dB. The chosen attention placement improves all three metrics.

The results does not follow the usual expectation that the better visual quality reduces the PSNR value, in this setup, both visual quality and pixel accuracy improved together.

Among all the evaluation metrics, FID shows the most sensitive metric in the comparison, It ranges from 7.42 to 10.65 across the variants, while SSIM only ranges from 0.695 to 0.716, and PSNR changes by 0.87 dB (23.98–24.85 dB). This can be interpreted to FID capturing differences in overall structure that are less visible in the other metrics. Based on this findings, FID is used as the main metric for choosing the model for comparative analysis since overall realism and consistency matter in super-resolution models.

4.5.4 Selection of the Optimal R3GAN Variant

Based on the Metric Trade-off Analysis and Decision Criteria, SR-SAR3GAN-2 is selected as the best model among the variants based on training behavior, efficiency, and evaluation results. In of the efficiency, , SR-SAR3GAN-2 was able to reach an FID of less than 10 in about 3.5 days, while SR-SAR3GAN- 1 took around 9.5 days, Although Variant 1 was affected by interruptions and shared GPU use, Variant 2 still shows more stable progress under similar conditions.

SR-SAR3GAN-2 also continues to improve later in the training, Its FID drops from 7.95 at 70,000 steps to 7.42 at the end, while other variants level off around 40,000–50,000 steps. The placement of the attention block increased the model’s final performance and not just how fast it learns.

The decision of chosing SR-SAR3GAN-2 is also supported by its gradient penalty

behavior, Variant 2 stays within a stable range ($R1/R2 \approx 0.0007-0.0012$), with occasional spikes up to $R1 = 0.0053$ during harder updates.

4.6 Comparative Evaluation Results

This section compares the selected SR-SAR₃GAN variant with state-of-the-art super-resolution models. The comparison is not meant to prove that the proposed model is better than all existing methods. It checks whether the model performs within a comparable range under the same training conditions.

For a fair comparison, non-pretrained versions of REAL-ESRGAN and SWINIR were trained in the same environment and under the same resource limits as SR-SAR₃GAN.

4.6.1 Metric Results and Analysis

One sub-hypothesis of this study states that an SR-SAR₃GAN variant can perform comparably to state-of-the-art super-resolution models (REAL-ESRGAN and SWINIR) in both perceptual and objective evaluation.

Table 4.3. Objective metrics (FID, SSIM, PSNR) and survey-based MOS.

SR Model	↓ FID	SSIM ↑	PSNR ↑	MOS
REAL-ESRGAN	1.58	0.84	29.23	2.55
SR-SAR ₃ GAN-2	7.42	0.72	24.85	3.94
SWINIR	26.20	0.86	30.53	2.88

The objective metrics and survey scores do not follow the same ranking. SR-SAR₃GAN-2 gets the highest MOS score, although its objective metric scores are not the highest. REAL-ESRGAN has the best FID, while SWINIR has higher SSIM and PSNR. Human viewers still rate SR-SAR₃GAN-2 higher.

One possible reason is the way SR-SAR₃GAN-2 produces facial outputs. It tends to favor coherent facial features over exact pixel matching. The output may not match the ground truth pixels as closely, which lowers PSNR and SSIM. Even so, the images

can look more natural to viewers, leading to a higher MOS.

4.6.2 Blind Ranking Evaluation

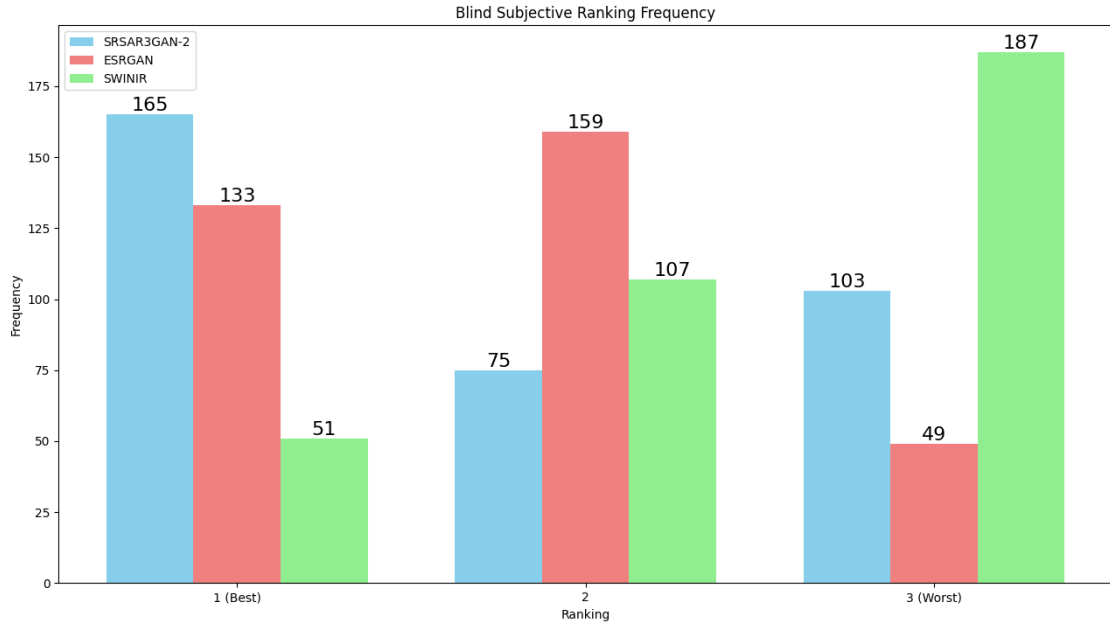


Figure 4.12. Summary of blind ranking evaluation.

The blind ranking results follow the same pattern as the MOS results. SR-SAR₃GAN-2 has the highest number of best ranks, but it also receives a moderate number of worst ranks. It is often preferred, but not in every case.

REAL-ESRGAN has a more balanced distribution, with more neutral rankings and fewer extreme ratings. Its outputs are stable but less noticeable to participants.

SWINIR has the highest number of worst ranks (187) and the lowest number of best ranks (51). Even with good reconstruction metrics, it receives lower preference in the blind ranking.



Figure 4.13. Sample survey outputs: SR-SAR3GAN-2, Real-ESRGAN, SwinIR.

SR-SAR3GAN-2 records 8.1% best, 21.9% neutral, and 30.0% worst rankings. The responses are mixed. In many cases, participants prefer its outputs, but in some cases they rate them poorly. Artifacts or inconsistencies in some generated images may explain these lower ratings.

Its 165 best-rank assignments exceed REAL-ESRGAN by 24% and SWINIR by 223%. The pattern matches the MOS result, where SR-SAR3GAN-2 also receives the highest score.

4.6.3 Statistical Analysis

Table 4.4. Friedman and Wilcoxon tests for perceptual ranking results.

Test	Comparison	Statistic	p-value
Friedman	All models	7.6	0.02
Wilcoxon signed-rank	SR-SAR3GAN-2 vs REAL-ESRGAN	6.5	1.00
Wilcoxon signed-rank	SR-SAR3GAN-2 vs SWINIR	0	0.06

The Friedman test shows differences among the models ($\chi^2 = 7.6, p = 0.02$), so the perceptual rankings are not the same across all outputs. The post-hoc results place SR-SAR3GAN-2 statistically equivalent to REAL-ESRGAN ($p = 1.00$). It is slightly below SWINIR, but the difference is not statistically significant ($p = 0.06$). Based on this, SR-SAR3GAN-2 is within a comparable perceptual range to the reference models.

Table 4.5. Summary of SR model experiments.

Hypothesis Component	Evaluation Basis	Key Evidence	Result
Impact of self-attention placement on SR model	Ablation study across R3GAN variants	Distinct differences observed in FID, SSIM, and PSNR across attention placement strategies; mid-level attention preserved global structure while enhancing perceptual quality	Confirmed
Multi-metric comparison between SR-SAR3GAN-2 and baseline across identical training conditions	SR-SAR3GAN-2 achieved lower FID and higher SSIM and PSNR compared to baseline R3GAN under identical training conditions	SR-SAR3GAN-2 achieved lower FID and higher SSIM and PSNR compared to baseline R3GAN	Confirmed
Comparative analysis using objective metrics and MOS	SR-SAR3GAN-2 achieved perceptual quality comparable to REAL-ESRGAN and SWINIR trained on the same environment	Higher MOS indicating favorable human perception	Confirmed
Friedman test + Wilcoxon signed-rank tests	Friedman test indicated differences; pairwise Wilcoxon tests showed no significant difference between SR-SAR3GAN-2 and REAL-ESRGAN or SWINIR ($p \geq 0.05$)	Statistical tests confirm human-perceived quality comparability	Confirmed
Training stability and resource analysis	Stable convergence under single-GPU constraints, without collapse or excessive resource demand	Training completed successfully on a single GPU	Confirmed

CHAPTER 5. SUMMARY

5.1 Summary of Contributions

This study shows that R3GAN can be used beyond unconditional image generation, serving as a baseline for single-image super-resolution. By adapting R3GAN into a Super-Resolution GAN and adding self-attention through non-local blocks at different layers, the study evaluates its suitability for SR and the effect of attention placement on performance.

The baseline R3GAN achieved stable adversarial training with R1/R2 gradient regularization over 62,500 steps, providing a consistent reference for all variants. Strategic mid-level self-attention placement in SR-SAR3GAN-2 yielded the strongest overall performance, achieving a 30.3% FID improvement over baseline, while comparisons across variants confirmed that placement strategy critically determines effectiveness. Despite its smaller parameter footprint, SR-SAR3GAN-2 matched state-of-the-art perceptual quality against REAL-ESRGAN and SWINIR under identical training conditions, and earned the highest Mean Opinion Score among all evaluated models, revealing a notable divergence between human-perceived quality and pixel-level metrics. Finally, all variants achieved stable convergence on a single virtualized V100 GPU, establishing that competitive super-resolution GANs can be developed without multi-GPU clusters or bare-metal infrastructure.

5.2 Findings

1. R3GAN is a viable and stable baseline for single-image super-resolution achieving FID score of 10.65 without self-attention. The R1 and R2 gradient penalties maintain adversarial balance at 0.0016-0.0020, preventing mode collapse and enabling

stable training across all model variants, thus supporting the claims of Huang et. al. for the possibility of making R3GAN as a backbone for future GANs.

2. Self-attention placement has a significant impact on model performance. With different variants of the model improving from the baseline SR-SAR3GAN-1 showing 18.5% FID improvement over baseline, 30.3% for SR-SAR3GAN-2 and 3.9% FID improvement for SR-SAR3GAN-3. The performance differences across variants confirm that attention placement strongly affects training stability and output quality, with mid-level placement yielding the most favorable results.
3. Perceptual and objective metrics capture different aspects of quality. This is shown in the results of SR-SAR3GAN-2 achieved highest MOS (3.94) despite middle-tier PSNR (24.85 dB) and REAL-ESRGAN achieved highest PSNR (29.23 dB) but lowest MOS (2.55). FID showed 44% variance across models (1.58-26.2) vs PSNR's 3.5% variance (24.85-30.53 dB). Discrepancies between FID/MOS and PSNR/SSIM indicate that perceptual realism and pixel-level accuracy do not always align in GAN-based super-resolution.
4. SR-SAR3GAN-2 achieves perceptually comparable performance to state-of-the-art models, with subjective evaluations and statistical tests showing image quality on par with REAL-ESRGAN (Wilcoxon $p=1.00$) and marginally superior to SWINIR (Wilcoxon $p=0.06$) under identical training conditions. It earned 24% more first-place blind rankings than REAL-ESRGAN (165 vs. 133) while reaching this quality 45% faster, converging to $FID < 10$ in 3.5 days compared to 6.4.
5. Training advanced SR models is feasible in a single-GPU virtualized environment, as demonstrated by successfully training a 2.31M-parameter GAN on a single V100 (32GB VRAM) to a competitive FID of 7.42 in 11 days wall-clock time. Despite a 5–15% virtualization overhead, the setup achieved stable convergence while

sustaining near-full resource utilization across GPU (100%), CPU (80%), and RAM (97%, 31/32 GB).

5.3 Limitations

This study was conducted under several constraints that should be considered when interpreting the results. All models were trained in a virtualized single-GPU environment, which limited batch size, training duration, and architectural exploration, and may affect scalability compared to multi-GPU settings. The use of a modified FFHQ dataset limits the study to facial images, reducing generalization to other super-resolution domains. No dataset split was used during training, as models were trained on a fixed partition, which limits evaluation on unseen data.

Only a limited set of self-attention placements was tested, and transformer-based architectures were not explored, leaving other attention designs unexamined. Although MOS and blind ranking provide perceptual insight, they are subject to human bias and viewing conditions despite statistical checks. The evaluation also relies on FID, SSIM, and PSNR; adding other perceptual or task-based metrics could improve analysis.

Finally, occasional training interruptions and checkpoint resuming, especially for some variants, may have affected training efficiency and wall-clock time comparisons.

5.4 Recommendations

Future work should expand dataset diversity and improve architectural robustness of SR-SAR3GAN. Although this study focuses on facial images, future evaluations should test cross-domain performance using datasets of natural scenes and urban environments, with proper train, validation, and test splits.

Architecturally, future work should explore transformer-based methods and multi-head attention to better model hierarchical features. Further study of self-attention at different resolutions and hybrid attention designs may improve feature extraction and texture detail.

To address computational limits and improve training efficiency, future implementations should consider multi-GPU distributed training. This infrastructure upgrade would facilitate larger batch sizes, enable the exploration of deeper architectures, and significantly reduce training duration.

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APPENDICES

Appendix A. Overall Similarity Result


Page 2 of 4 - Integrity Overview
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Appendix B. AI Detection Result


Page 2 of 77 - AI Writing Overview
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Appendix C. Deed of Assignment

CARAGA STATE UNIVERSITY

DEED OF ASSIGNMENT

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WHEREAS, the material entitled Evaluating Self-Attention Placement in R3GAN for Fixed-Scale Single-Image Super-Resolution: Ablation and Comparative Analysis in Virtualized Single-GPU Environment authored by Kyr Djan R. Marimon and co-authored by Lyndon King L. Obenza with its principal office address at the Ampayon, Butuan City, hereafter referred to as Assignor, has ownership and made a copyright registration for his/her/their work

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Title of Work

WHEREAS, **CARAGA STATE UNIVERSITY** with principal address at Ampayon, Butuan City, represented by its President, **ROLYN C. DAGUIL, Ph.D.**, hereinafter referred to as Assignee, is desirous of acquiring interest therein.

NOW, THEREFORE, Assignor herein, by these presents do hereby assign and transfer unto said Assignee, the whole right and interest to the said copyright; the same to be held and enjoyed by Assignee hereof to the full end of the term for which said copyright as fully and entirely as the same would have been held by the Assignors herein had this assignment not been made.

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Before Me, a Notary Public for and in Butuan City, this ____ day of _____
 2024 personally appeared:

Competent Evidence of Identity (CEI)

ROLYN C. DAGUIL, Ph.D.

ID. NO. M98-009

KYR DJAN R. MARIMON

LYNDON KING L. OBENZA

, who

_____ is/are personally known to me
 _____ exhibited (a) competent evidence/s of identity/ies as stated above and
 known to me to be the same person/s who executed the foregoing instruments which
 consist of one (1) page sealed with my notarial seal and she/he/they acknowledged to me
 that his/her/their signature on the instrument was/were freely and voluntarily affixed by
 him/her/them for purposes stated therein.

He/She/They further declared that he/she/they has/have the authority to sign in
 behalf of the principal that he/she/they represent(s).

WITNESS MY HAND AND SEAL on the date and on the place first above
 mentioned.

DOC. NO. _____
 PAGE NO. _____
 BOOK NO. _____
 SERIES OF _____

Appendix D. Certification of Substantial Use of University Resources

 Caraga State University Technology Transfer and Licensing Office CSU-Main Campus, Ampayon, Butuan City Competence Service Uprightness www.itso.carsu.edu.ph	 
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Certification of Substantial Use of University Resources

This is to certify that aside from the ordinarily available resources of the University such as office, library, computers and storage servers during the course of the development of the research entitled _____,

I/we have utilized any of the following resources:

(Please check the appropriate boxes)

Laboratory Facilities:

☐ Experimental Apparatus
☐ Specialized Software (please specify):
☐ Lab Instruments
☐ Data Analysis Tools
☐ Technical Support (e.g. Training, Troubleshooting, etc.)
☐ Farm/Machine Shop
☐ Other (please specify): ICTC HPC

Funding Resources:

☐ Personal Funds/Resources
☐ Grants/Funding/Wages/Allowances/Stipend/Salary
☐ Scholarships
☐ Industry Partnerships
☐ Collaboration with Other Institutions
☐ Other (please specify):

☐ I/We have not utilized any of the resources mentioned above.

Remarks:


 Signature over Printed Name KYR DAN R. MARIMON


 LYNDON KING L. OBENZA

Date May 13, 2026

